## Appendix A. Supplementary Information

### *Appendix A.1. Study population and stratification*

The cohort includes registered inhabitants of The Netherlands born between 1930 and 2020. For inclusion in the cohort, we require that we can reliably track people in the main datasets from 2018 through 2022. This means that throughout that period, they must either be found in the municipal registration datasets every year, or be recorded as deceased. In the latter case, they must have a recorded place of death and cause of death. With these criteria, we limit censoring and missing data problems. Such problems could be related to immigration and emigration of people in that period, leading to unwanted outflux and influx of people in the cohort. Such problems could also be related to other people whose medical history, including death records, is poorly documented for other reasons. These can, for example, be people who become homeless during the study period, but also people whose accessible data becomes inaccessible due to other (such as privacy) reasons.

We also use a number of other exclusion criteria, related to excluding irrelevant deaths from before the main period of interest (which is 2021 and later), and related to excluding a few very small age groups or age groups with very low mortality. See Figure 1 in the main document for a Consort Flow Diagram of the study population and exclusions. The number of eligible persons in the Netherlands in 2018 that we start with is 16,931,761, and after applying the inclusion/exclusion criteria we are left with 15,915,865 persons.

We use stratification into smaller cohorts based on year of birth (e.g., 1930-1940: born in or after 1930, and before 1940), as opposed to age directly (e.g., 80-90 years). This has the advantage that, in an analysis of mortality over multiple years (2021 to 2023 in our case), there are no new people to enter each year because they turn 80 in that year, nor are people disappearing because they turn 90. In other words, we follow a fixed group of people selected based on year of birth over several years, with people disappearing only because of mortality. This is consistent with (most) analyses in [18]. For clarity and simplicity, we refer to these as age cohorts.

For the older groups, we use a 10-year interval (e.g., 1930-1940); for the relatively younger groups, we use a 20-year interval (e.g., 1980-2000). An age cohort with 20-year interval results in some loss of precise information: after all, there may be different mortality patterns among people of 1980-1990 compared with people of 1990-2000. We do this because the mortality rates among those relatively younger groups, especially when looking at mortality per 4 weeks (which we do in some analyses), is so low that data becomes very noisy and also regularly falls below 10 people. The CBS Microdata research platform data export rules do not allow exporting such data based on small numbers of people, due to the privacy-related risk of individual or group disclosure. When such low numbers do still occur despite our aggregation, in all reported and visualized data the numbers (and derived percentages) have been rounded down to zero. In summary, we use the following age cohorts in our main analyses:

1. 1930-1940
2. 1940-1950
3. 1950-1960
4. 1960-1980
5. 1980-2000
6. 2000-2020

Like [18], we use these age cohorts in combination with a distinction based on "WLZ" use, i.e. being registered in the CBS Microdata databases as a recipient or user of the Long-Term Care Act ("Wet Langdurige Zorg"). WLZ use is an important indicator of severe underlying vulnerability (e.g., nursing home admission) and it is therefore important to consider them as a separate subgroup. WLZ use is very rare in the youngest age cohort, 2000-2020; that subgroup was excluded from further analysis. In the oldest age cohorts, WLZ mostly concerns nursing and care homes for the elderly. Among younger groups, WLZ also occurs frequently, but more often due to severe congenital mental or physical disabilities. For

WLZ users, we know the "type" of care, related to type of disability or type of care home, as well as the "level" of care, which we use as a proxy for vulnerability.

For all age cohorts, we also calculate "normal" (baseline, expected) mortality, which is used for comparisons to measured mortality in the years 2021-2023 in some analyses. Normal mortality for an age cohort is the expected mortality rate for that age cohort over a year, as calculated based on 4-week averages in the pre-pandemic years 2016 through 2019, for an age cohort that was the same age in those earlier years. In those calculations, all individuals regardless of WLZ use were included. This means that realized mortality for a no-WLZ cohort should naturally be lower than that normal, baseline rate, and realized mortality for a WLZ cohort should naturally be higher.

*Appendix A.2. Vaccination status: definitions and missing data*

Most studies, i.e. both the clinical trials [22] and most observational studies including [18], use a period of two weeks after completion of the vaccination course (for the primary course this typically consists of two doses several weeks apart) before someone is considered to be (fully) vaccinated. This makes sense given the expected immunological effects of vaccination. However, for the purpose of assessing complete, real-world, all-cause morbidity and mortality effects in practice, it is reasonable and fair to start follow-up for vaccinated individuals at the moment of the first dose – which is what we do in this paper. This way, we can follow mortality patterns and differences immediately after vaccination, i.e. to also assess short-term effects, and avoid the type of biases identified in [29, 81, 32]. Follow-up of mortality was done until the end of 2023, allowing evaluation of longer-term effects of vaccination as well.

An important limitation of the central Dutch vaccination database CIMS that we and others (including [18]) use to determine vaccination status is that an imprecisely known percentage of the vaccinations is missing, mostly due to lack of consent from individuals or from certain institutions where vaccinations were administered. People for whom vaccinations are missing in the central vaccination databases are necessarily considered "unvaccinated". Previous reports estimate approx. 7% missing and suggest this has little impact on the analyses [18]. However, recent reports [24, 19, 31] suggest, based in part on checks against independent vaccination data, that this percentage may be significantly higher, at least for certain subgroups, and unevenly distributed among subgroups. There is currently no precise information on this.

*Appendix A.3. Covid-19 and other causes of death definitions*

Covid mortality is defined by the person having an official registered cause of death of Covid-19 based on ICD10 codes U07.1 and U07.2, using the available central CBS causes-of-death database. Non-Covid mortality corresponds to any other cause of death.

It is well-known that this type of cause of death determination and delineation is not very precise. There may be either over- or under-attribution of deaths to Covid. A Covid death typically occurs in individuals with various pre-existing comorbidities and vulnerability; and there may be biases due to differential attribution of Covid as the cause of death depending on testing protocols, age groups, vaccination status, and potentially other factors. In the Netherlands, as in many countries, the WHO guidelines [39] were used for Covid death definition during the pandemic. These guidelines are generally recognized as being somewhat biased toward over-attribution to Covid rather than under-attribution [40], given their broad and liberal definition of a Covid death ("a death resulting from a clinically compatible illness, in a probable or confirmed COVID-19 case, unless there is a clear alternative cause of death that cannot be related to COVID disease (e.g. trauma)"). Furthermore, during the pandemic there was widespread testing for people in hospitals, nursing homes, and general practitioner care, making it unlikely that a Covid-19 diagnosis shortly before death was missed. On the other hand, although the CBS research platform does have a dataset containing centrally registered Covid-19 diagnostic test results from several sources, that dataset is relatively sparse and does not provide accurate or complete information on previous SARS-CoV2 infections. Thus, given the limitations of these population-level datasets, it is impossible to accurately determine the role of post-acute sequelae of infection in the cause of death, and it is similarly impossible to stratify or analyze cohorts based on complete SARS-CoV2 infection history.

Regardless of all the complexities and uncertainties surrounding Covid-19 as a cause of death, this definition derived from the centrally and officially registered causes-of-death dataset is the best that can be done given the databases in this central Dutch CBS research platform and the lack of detailed patient and testing data therein. It is the same approach as used in [18, 19] and most studies on Covid mortality, and it is at least highly *indicative* of actual Covid vs. non-Covid mortality. In any case, as already mentioned, we primarily focus on all-cause mortality, for which such cause of death attribution does not play a role.

In some analyses (in particular the calendar time analyses), we also look in more detail at various other causes of deaths. We use categories of causes of death, denoted by letters, derived from ICD10 codes:

1. U: Covid-19 (U07.1 and U07.2);
2. I/R: cardiovascular causes of death (I) and R9 ("Other sudden cardiac arrest death, cause unknown" etc.);
3. C/D: cancer (C01-C99, D01-D49);
4. Other: other or unknown.

*Appendix A.4. Vaccine effectiveness (VE) calculations*

To calculate vaccine effectiveness (VE), we use standard formulas based on directly measured data on the number of deaths in vaccinated and unvaccinated groups in a given measurement period. The standard formula used for (relative) VE is:

$$VE_{RRR} = (1 - \frac{p_v}{p_u}) \times 100 \tag{A.1}$$

in which:

1. $VE_{RRR}$ represents vaccine effectiveness in terms of Relative Risk Reduction (RRR), expressed as a percentage;
2. $p_u$ is the proportion (rate) of people in the unvaccinated group who die during the measurement period;
3. $p_v$ is the proportion of people in the vaccinated group who die during the measurement period.

In addition, we also look at Absolute Risk Reduction (ARR): what is the absolute risk of disease itself, or dying, for a given group, and to what extent is that absolute risk reduced by the vaccine? This provides complementary insight into the extent to which vaccine protection is relevant and substantial; and it provides additional opportunities for sanity checks, as done in this paper. The formula to calculate $VE_{ARR}$ simply is:

$$VE_{ARR} = (p_u - p_v) \times 100 \tag{A.2}$$

*Appendix A.5. Matching algorithm*

We use a fixed cohort, "never-treated" (unvaccinated) vs. "ever-treated" (vaccinated) design [41, 42, 43] together with matching. Our matching algorithm matches vaccinated and unvaccinated individuals, within each single age cohort, based on age (per 5 years, i.e. more fine-grained than realized through the age stratification itself), sex, and comorbidity information. See 2 in the main document for a Consort Flow Diagram of the stratification and matching process. Thus, in this design unvaccinated individuals are defined as individuals who, according to vaccination database CIMS, do not receive any Covid vaccination dose. (Below, and in Appendix A.2, we discuss alternative definitions and approaches.) Since there are many more vaccinated than unvaccinated people in almost all age cohorts (apart from the very youngest), we use a 3:1 ratio in matching vaccinated vs. unvaccinated. In many (esp. older) age cohorts, the actual ratio of vaccinated vs. unvaccinated is higher than that, since vaccination rate in such cohorts was often around 90% (see Table 1). However, the choice for this 3:1 ratio was made in order to make the best use

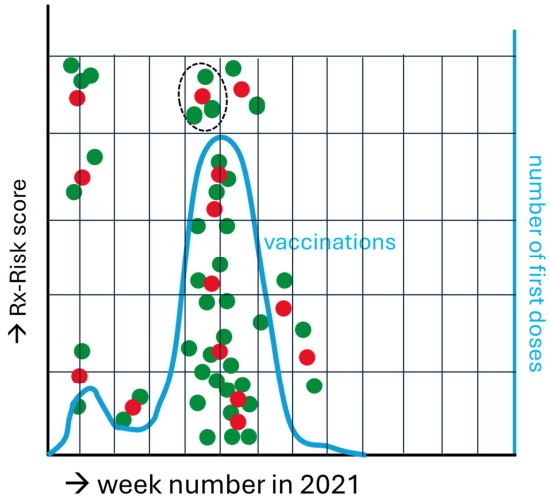



Figure A.1: Schematic illustration of how matching is done; based on comorbidities (vertical axis) and start date (horizontal axis), using a ratio of 3:1. An individual person is denoted by a circle, with green indicating a vaccinated person and red indicating an unvaccinated person. The thin grid lines represent the "Coarsened Bins" of CEM matching.

of the available data on the more 'rare' unvaccinated individuals (i.e., to lose as few of them as possible in the matching process).

As for the comorbidity information, in preliminary analyses we experimented with using hospital diagnosis data for this (similar to [18]), using the well-known Charlson and Elixhauser Comorbidity Index metrics. However, those preliminary analyses indicated that comorbidity metrics based on medication use worked better to predict mortality.

Medication use per individual person is available in the CBS database at the per-year level (i.e. which medications an individual has had in a particular calendar year, as reimbursed through national insurance). It is encoded using the standard ATC coding [45], truncated to the first four characters. We use the medication data of the previous year before the year of vaccination, i.e. 2020, to avoid confounding with disease occurring in 2021, possibly after vaccination or infection in 2021. From this data, we calculate the weighted medication-based Rx-Risk Comorbidity Index [45] as the main comorbidity metric. A person may have medications in multiple ATC categories, and based on the Rx-Risk Comorbidity table and weights, corresponding points are added up to arrive at the total score.

For people receiving long-term care (WLZ), frequently medication data is missing, presumably due to their receiving medication through other, WLZ-related insurance channels not provided to the central medication database. However, at the same time for them a very useful proxy for comorbidity level is available: WLZ *type* and WLZ *level of care*, which codes for the "Care Severity Package". Thus, for WLZ recipients that information is used for matching instead of Rx-Risk.

Starting point (start date) of follow-up for vaccinated people is the date of their first vaccination dose. Thus, we know that those people are alive at that time. Our matching algorithm requires that an unvaccinated person to be matched with a vaccinated person is also alive on the vaccination date of that vaccinated person, in line with the well-known prescription time-distribution matching (PTDM) approach [42]. Doing so aims to prevent the well-known problem of immortal time bias [28, 42] and seasonality-related bias in this type of retrospective observational research, which may have affected earlier analyses [18] (although their methods descriptions are insufficiently clear to us to determine this exactly). After all, if we do *not* do this correction and choose January 1, 2021 as the start date of follow-up for everyone, as in [18], then the vaccinated may have an 'unfair advantage'. This is because we know that they were at least alive on their vaccination date (often in the spring of 2021); whereas unvaccinated people may have

died earlier, in the first winter months of 2021 which coincided with one of the strongest Covid-19 waves, thus contributing to mortality in the unvaccinated group. However, bias cannot fully be eliminated (there is no perfect way to do this), esp. confounding-by-indication bias, as the fact that someone has not been vaccinated may in itself be an important indication of high vulnerability or of having died coincidentally just before they would be vaccinated. Appendix A.11 provides some extra insight into this remaining bias.

Other ways than ours to select and investigate unvaccinated versus vaccinated populations exist, such as including unvaccinated person days until the person is vaccinated [88]. However, such approaches have other, and in our view more important, disadvantages. Such approaches can make sense when vaccinated and unvaccinated groups are relatively well-balanced and vaccination moments are more or less randomly distributed over the year. However, during the Covid-19 vaccination program, within a typical stratified age group, most people (around 90%) started out unvaccinated and became vaccinated in the space of a few weeks. Thus, this alternative approach would lead to the 'unvaccinated' control group to be made up of a very heterogeneous mix of people: people who receive no vaccination at all (for whatever reason), together with (many more) people who *are* vaccinated but just a little later than other vaccinated people. In one such approach, the latter subgroup would all be censored at the time of vaccination, and thus have only very short follow-up times. That heavy censoring would be *informative* censoring – highly undesirable and leading to unpredictable bias. This is because those censored subjects come only (or primarily) from that latter subgroup, leaving the study at a moment when they are healthy enough for vaccination, and increasingly leaving behind in the 'unvaccinated' cohort the first subgroup of truly and intentionally unvaccinated people, a subgroup which in all likelihood is different in some respects. In yet another approach, vaccination status would be a time-dependent variable, e.g. in a Cox Proportional Hazards regression approach. This leads to substantial statistical issues that are very difficult to measure and correct for (as seen in [88]), related to the immense and unbalanced heterogeneity of the 'unvaccinated' subjects, most of which cross over to being vaccinated within a very short timeframe, while only a limited few, in fact those of most interest, do not. All in all, we believe that our matching and selection and analysis approach is both more natural and less prone to undesirable biases and statistical issues.

For our specific matching algorithm, we use the *Coarsened Exact Matching* (CEM) algorithm [46]. This is less commonly used in epidemiological studies than the Propensity Score Matching (PSM) algorithm [44], but still widely used, and in our opinion more suitable for our current use case. In Coarsened Exact Matching, all variables to be matched are classified into a limited number of "coarsened" bins (e.g., low/medium/high = 0/1/2, for a continuous variable). Subsequently, they are combined into a vector of discrete-value bin variables that gives a simplified but informative characterization of the person. Matching requires that the treatment (vaccinated) and control (unvaccinated) person have the "exact" same CEM vector values. Figure A.1 schematically illustrates the CEM-based matching process.

In PSM, on the other hand, an additional, separate regression model is created that 'predicts' the probability for a person to be in the treatment group (in our case: probability to receive vaccination as a function of age, gender and comorbidities) based on the available data in the dataset. For each person who is in the treatment (vaccination) group and has a certain PSM score, in the matching process we choose a control (unvaccinated) person with (almost) the same PSM score.

The PSM approach is appropriate, widely used, and sensible for situations where one is doing a study of, for example, a particular heart surgery (treatment), and one wants to match treatment and control individuals based on their 'eligibility for this type of heart surgery'—which depends on age and comorbidities etc., and is usually determined by a physician. But in the case of Covid vaccinations, social and psychological factors independent of health played a very large role in this choice, and with that the probability of being vaccinated. As a result, it may very well be the case that a young healthy 25 year old woman has the same PSM score as an old, obese, 70 year old man, who perhaps may come from a very different socio-economic background. At the same time, their probabilities of dying from Covid-19, or from any condition, are very different. Thus, according to this logic they should not be matched—no sensible balancing follows from this, in terms of a priori probability of dying. CEM matching does not

have this problem and is therefore the better choice here, in our opinion.

Before matching using our CEM algorithm, all potential matches are randomly shuffled into a list. Next, starting from the top of the list, vaccinated individuals and unvaccinated individuals are matched based on their CEM vector and the vaccination date (rounded to week number for the three vaccinated matches; and using the 'alive on that date' criterion for the unvaccinated match). This process continues until no more valid matches (using the 3:1 vaccinated/unvaccinated ratio) are found. Individuals for whom no matches are found are excluded from the analysis. This is as it is intended: those individuals are either outliers that, if included, would bias the subsequent analysis too much; or they are excluded so as to preserve the balanced 3:1 vaccinated/unvaccinated ratio.

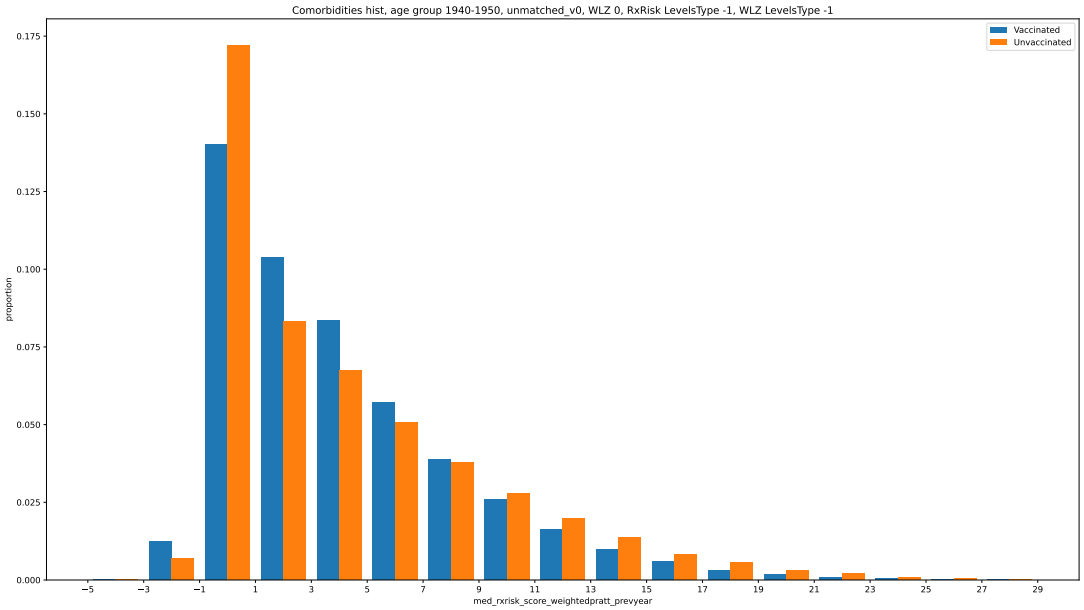



Figure A.2: Distribution of Rx-Risk comorbidity scores *before* CEM matching, for the 1940-1950 age group, no WLZ.

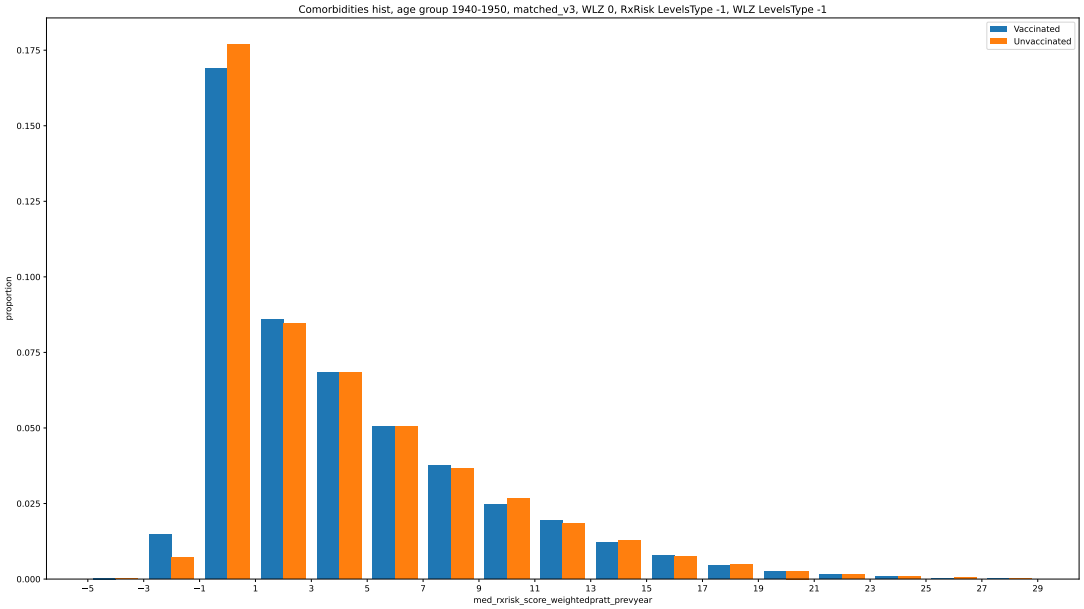



Figure A.3: Distribution of Rx-Risk comorbidity scores *after* CEM matching, i.e. for the *matched* 1940-1950 age group, no WLZ.

Figures A.2 and A.3 show the distribution of Rx-Risk comorbidity scores before and after CEM matching, respectively, for the 1940-1950 birth year group, no WLZ. A first observation is that most people have very low Rx-Risk scores (bars on the left side of each graph). A small proportion have higher values. This indicates more medication use, and medication use for more serious diseases; and thus higher vulnerability. Only a very small proportion has very high values ($> 10$); but it is expected that relatively high mortality does come from that very small proportion. In other words, the relationship of Rx-Risk to probability of death is expected to be highly nonlinear.

Importantly, we see that before matching, in the unvaccinated group more people have such very high Rx-Risk values than in the vaccinated group. This indicates a potential healthy vaccinee effect (at least before matching). However, the differences do not appear to be large: variation within the vaccinated or the unvaccinated group is much larger than the difference between the two groups. In any case, after matching (Figure A.3) we see that the groups are well-balanced. We obtain similar results in other age cohorts. We also see similar, well-balanced results on the other matching variables (gender, age; and for WLZ recipients: WLZ type and level). In the older WLZ (long-term care) groups, after matching, measured Rx-Risk comorbidity was somewhat higher in the unvaccinated group. As mentioned, Rx-Risk was not used for matching in those groups as medication use data in those groups is more often incomplete; but this may still indicate some real imbalance.

We conclude that the matching process has worked as designed, and at least based on these comorbidity variables, has made the unvaccinated and vaccinated groups more similar and comparable. Appendix A.6 provides basic descriptive data of the matched groups for the various age and WLZ groups.

*Appendix A.6. Basic descriptive statistics data for the matched groups*

Tables A.1 and A.2 show basic descriptive statistics for the stratified age/WLZ cohorts after matching, including number of people per group, percentage male, and mean Rx-Risk comorbidity. As described above, a 3:1 matching ratio was used to match vaccinated to unvaccinated individuals. As in most analyses in this study, the youngest (2000-2020) age group with WLZ (long-term care recipients) is discarded due to insufficient amount of data. For the other WLZ cohorts, as described above (see Appendix A.5) WLZ "type" and "level" of care were used as a proxy for comorbidity in matching instead of Rx-Risk (which for WLZ users is often missing). This explains why there are relatively large residual differences (Standardized Mean Differences) in mean Rx-Risk score for vaccinated vs. unvaccinated WLZ recipients, despite matching; and it suggests differences and potential imperfections in the various comorbidity (proxy) metrics and matching based on them, for WLZ recipients. As can also be seen in Table A.2, among WLZ long-term care recipients, in the older groups the most common WLZ type is 1, corresponding to nursing homes and assisted living facilities. In the younger groups, the most common WLZ type is 4, corresponding to care for individuals with congenital intellectual disabilities.

Table A.1: Baseline characteristics of the matched cohorts, no long-term care (WLZ = 0); with Standardized Mean Difference (SMD) where appropriate.

| **Age cohort:** | **WLZ: 0** | **Status Vaccinated** | **Status Unvaccinated** | **SMD** |
|---|---|---|---|---|
| **1930-1940** | | | | |
| Number of people | N (%) | 22524 (75.0) | 45401 (25.0) | |
| Male | N (%) | 56028 (41.1) | 18676 (41.1) | 0.000 |
| Age | mean (std.dev.) | 85.6 (2.74) | 85.7 (2.76) | -0.024 |
| Rx-Risk Comorbidity | mean (std.dev.) | 5.60 (5.40) | 5.69 (5.39) | -0.017 |
| **1940-1950** | | | | |
| Number of people | N (%) | 299337 (75.0) | 99779 (25.0) | |
| Male | N (%) | 137430 (45.9) | 45810 (45.9) | 0.000 |
| Age | mean (std.dev.) | 75.9 (2.79) | 75.9 (2.82) | 0.009 |
| Rx-Risk Comorbidity | mean (std.dev.) | 3.56 (4.73) | 3.65 (4.69) | -0.021 |
| **1950-1960** | | | | |
| Number of people | N (%) | 517875 (75.0) | 172625 (25.0) | |
| Male | N (%) | 240441 (46.4) | 80147 (46.4) | 0.000 |
| Age | mean (std.dev.) | 66.1 (2.86) | 66.1 (2.86) | 0.017 |
| Rx-Risk Comorbidity | mean (std.dev.) | 2.35 (4.09) | 2.44 (4.04) | -0.023 |
| **1960-1980** | | | | |
| Number of people | N (%) | 1748340 (75.0) | 582780 (25.0) | |
| Male | N (%) | 908703 (52.0) | 302901 (52.0) | 0.000 |
| Age | mean (std.dev.) | 51.8 (5.64) | 51.7 (5.63) | 0.016 |
| Rx-Risk Comorbidity | mean (std.dev.) | 0.80 (2.85) | 0.86 (2.84) | -0.021 |
| **1980-2000** | | | | |
| Number of people | N (%) | 2765700 (75.0) | 921900 (25.0) | |
| Male | N (%) | 1396077 (50.5) | 465359 (50.5) | 0.000 |
| Age | mean (std.dev.) | 51.8 (5.64) | 51.7 (5.63) | 0.006 |
| Rx-Risk Comorbidity | mean (std.dev.) | 0.57 (2.10) | 0.60 (2.11) | -0.014 |
| **2000-2020** | | | | |
| Number of people | N (%) | 1339038 (75.0) | 446346 (25.0) | |
| Male | N (%) | 679686 (50.7) | 226562 (50.7) | 0.000 |
| Age | mean (std.dev.) | 51.8 (5.64) | 51.7 (5.63) | 0.050 |
| Rx-Risk Comorbidity | mean (std.dev.) | 0.19 (1.28) | 0.21 (1.28) | -0.015 |

In general, when comparing these descriptive statistics in the matched groups to the overall population descriptive statistics displayed in Table 1 (main document), the values suggest that the matching algorithm delivers results that are well-representative of the overall study population and well-matched.

Table A.2: Baseline characteristics of the matched cohorts, with long-term care (WLZ = 1); with Standardized Mean Difference (SMD) where appropriate.

| **Age cohort:** | **WLZ: 1** | **Status Vaccinated** | **Status Unvaccinated** | **SMD** |
|---|---|---|---|---|
| **1930-1940** | | | | |
| Number of people | N (%) | 22524 (75.0) | 7508 (25.0) | |
| Male | N (%) | 6723 (29.8) | 2241 (29.8) | 0.000 |
| Age | mean (std.dev.) | 86.84 (2.82) | 86.85 (2.80) | -0.003 |
| Rx-Risk Comorbidity | mean (std.dev.) | 5.16 (6.40) | 6.58 (6.91) | -0.212 |
| WLZ most common type | mode | 1 | 1 | 0.000 |
| N of WLZ most common type | N (%) | 22053 (97.9) | 7351 (97.9) | 0.000 |
| WLZ level | mean (std.dev.) | 5.04 (1.01) | 5.04 (1.01) | 0.000 |
| **1940-1950** | | | | |
| Number of people | N (%) | 11223 (75.0) | 3741 (25.0) | |
| Male | N (%) | 4908 (43.7) | 1636 (43.7) | 0.000 |
| Age | mean (std.dev.) | 77.1 (2.86) | 77.1 (2.87) | 0.001 |
| Rx-Risk Comorbidity | mean (std.dev.) | 4.55 (6.25) | 5.60 (6.73) | -0.163 |
| WLZ most common type | mode | 1 | 1 | 0.000 |
| N of WLZ most common type | N (%) | 9660 (86.1) | 3220 (86.1) | 0.000 |
| WLZ level | mean (std.dev.) | 5.25 (1.18) | 5.25 (1.18 | 0.000 |
| **1950-1960** | | | | |
| Number of people | N (%) | 6084 (75.0) | 2028 (25.0) | |
| Male | N (%) | 3276 (53.8) | 1092 (53.8) | 0.000 |
| Age | mean (std.dev.) | 66.8 (2.85) | 66.8 (2.92) | 0.006 |
| Rx-Risk Comorbidity | mean (std.dev.) | 3.25 (5.58) | 4.437 (6.26) | -0.188 |
| WLZ most common type | mode | 1 | 1 | 0.000 |
| N of WLZ most common type | N (%) | 3042 (50.0) | 1014 (50.0) | 0.000 |
| WLZ level | mean (std.dev.) | 5.43 (1.40) | 5.43 (1.40) | 0.000 |
| **1960-1980** | | | | |
| Number of people | N (%) | 8847 (75.0) | 2949 (25.0) | |
| Male | N (%) | 5172 (58.5) | 1724 (58.5) | 0.000 |
| Age | mean (std.dev.) | 52.2 (5.87) | 52.1 (5.92) | 0.011 |
| Rx-Risk Comorbidity | mean (std.dev.) | 2.13 (4.45) | 3.05 (5.15) | -0.191 |
| WLZ most common type | mode | 4 | 4 | 0.000 |
| N of WLZ most common type | N (%) | 5133 (58.0) | 1711 (58.0) | 0.000 |
| WLZ level | mean (std.dev.) | 5.23 (1.57) | 5.23 (1.57) | 0.000 |
| **1980-2000** | | | | |
| Number of people | N (%) | 21639 (75.0) | 7213 (25.0) | |
| Male | N (%) | 13356 (61.7) | 4452 (61.7) | 0.000 |
| Age | mean (std.dev.) | 29.8 (5.26) | 29.7 (5.22) | 0.017 |
| Rx-Risk Comorbidity | mean (std.dev.) | 1.43 (3.22) | 1.59 (3.38) | -0.048 |
| WLZ most common type | mode | 4 | 4 | 0.000 |
| N of WLZ most common type | N (%) | 19821 (91.6) | 6607 (91.6) | 0.000 |
| WLZ level | mean (std.dev.) | 4.87 (1.66) | 4.87 (1.66) | 0.000 |

*Appendix A.7. Additional Kaplan-Meier survival analysis data for various age cohorts*

Tables 2 and 3 in the main document show summary data derived from the Kaplan-Meier data calculated for all cohorts, based on All Cause Mortality data. Table 2 shows survival percentages of the unvaccinated ($S_u$) and vaccinated ($S_v$) cohorts at different moments (numbers of days) since start of follow-up. Table 3, in turn, shows VE values (ACM) as calculated directly from these KM survival data, for the individual periods defined by these follow-up moments. Due to the large systematic biases identified in this paper, and the corresponding violations of core statistical assumptions regarding homogeneity of research groups and proportional hazards (see 2.2 and 3.2), all these results should be interpreted with great caution. Here we present a number of additional figures displaying KM survival curves calculated for some of these age/WLZ cohorts (whose summary results are shown in those tables).

Figure A.4 shows Kaplan-Meier survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, WLZ (long-term care) recipients, for mortality from Covid; cause of death other than Covid; and ACM, respectively (see section 3.2 for explanation and interpretation). Here and in the other Kaplan-Meier figures in this paper where "confidence intervals" are shown around the survival curves themselves, these are the standard exponential (log-log) Greenwood confidence intervals used for KM curves. These do not take into account the large systematic biases, but are useful to represent uncertainty due to (low) number of people at risk and number of events.

At the time of data extraction and analysis, not all death records from final analysis year 2023 were complete, which explains the unusual flattening of survival curves at the very end of follow-up (an additional small artifact).

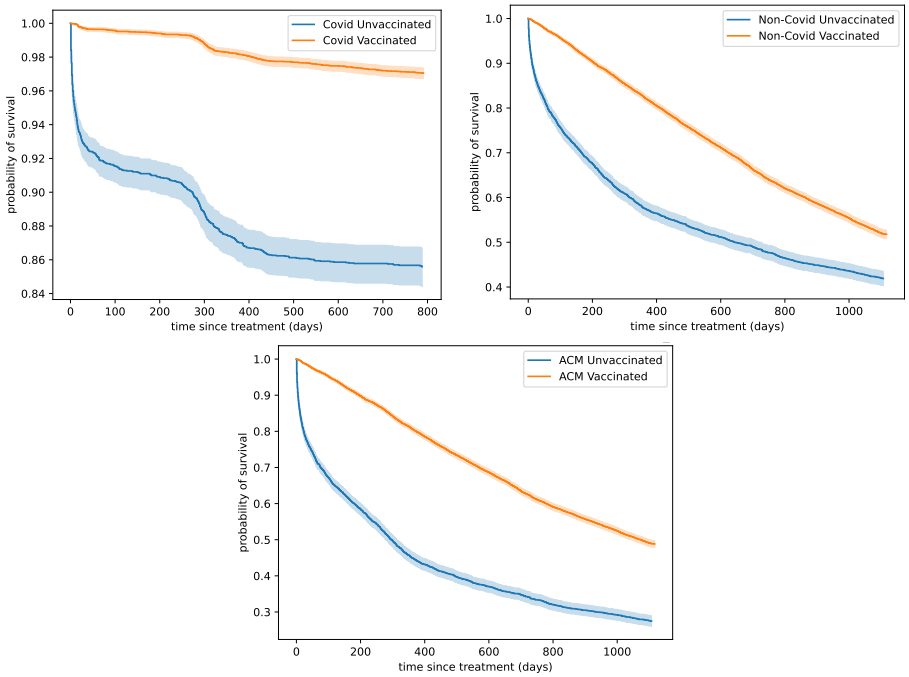



Figure A.4: Survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, *with* WLZ (long-term care), for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively.

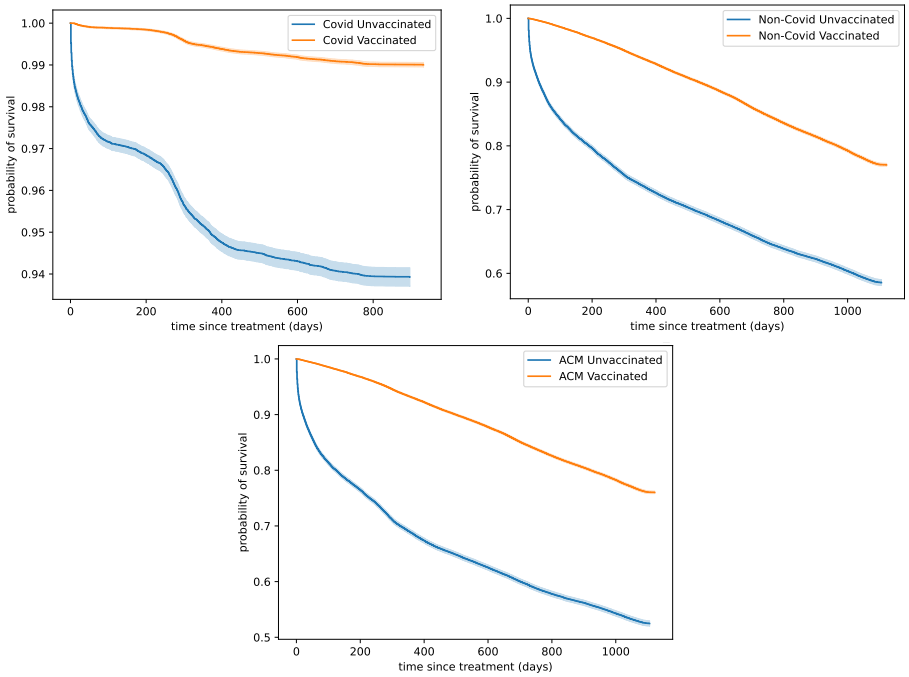



Figure A.5: Survival curves of vaccinated and unvaccinated individuals of age cohort 1930-1940, no WLZ, for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively.

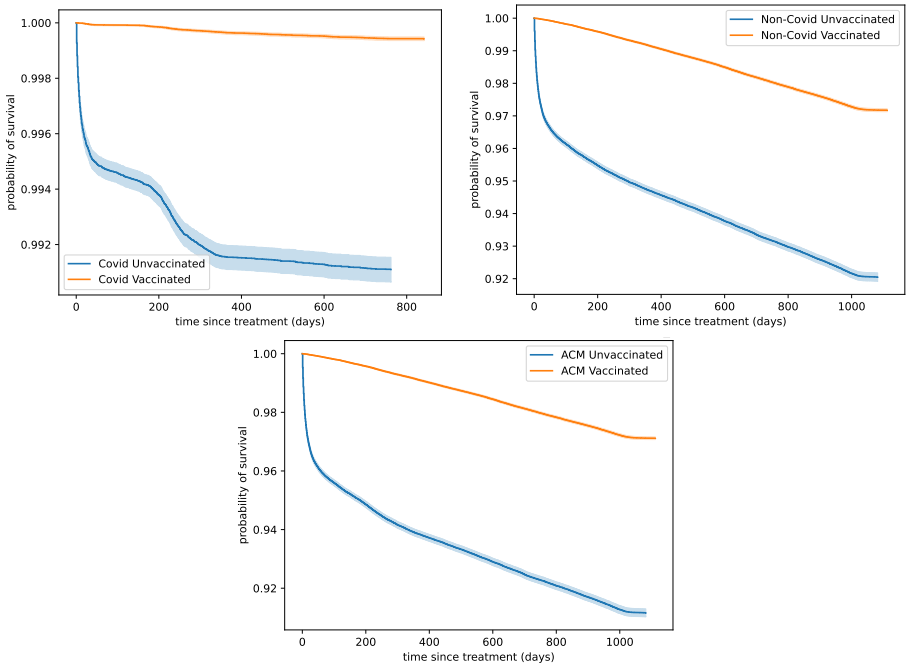



Figure A.6: Survival curves of vaccinated and unvaccinated individuals of age cohort 1950-1960, no WLZ, for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively.

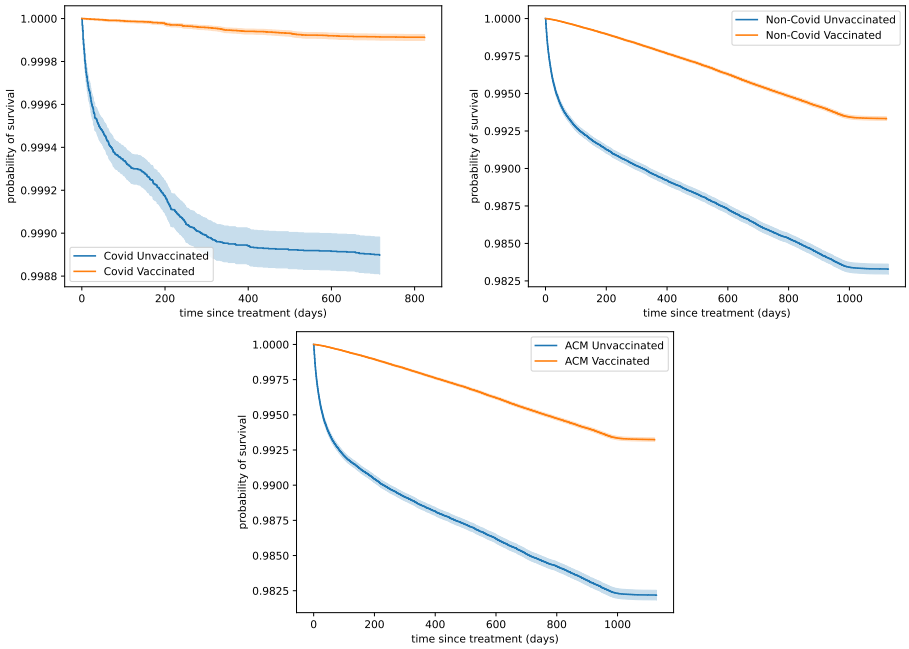



Figure A.7: Survival curves of vaccinated and unvaccinated individuals of age cohort 1960-1980, no WLZ, for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively.

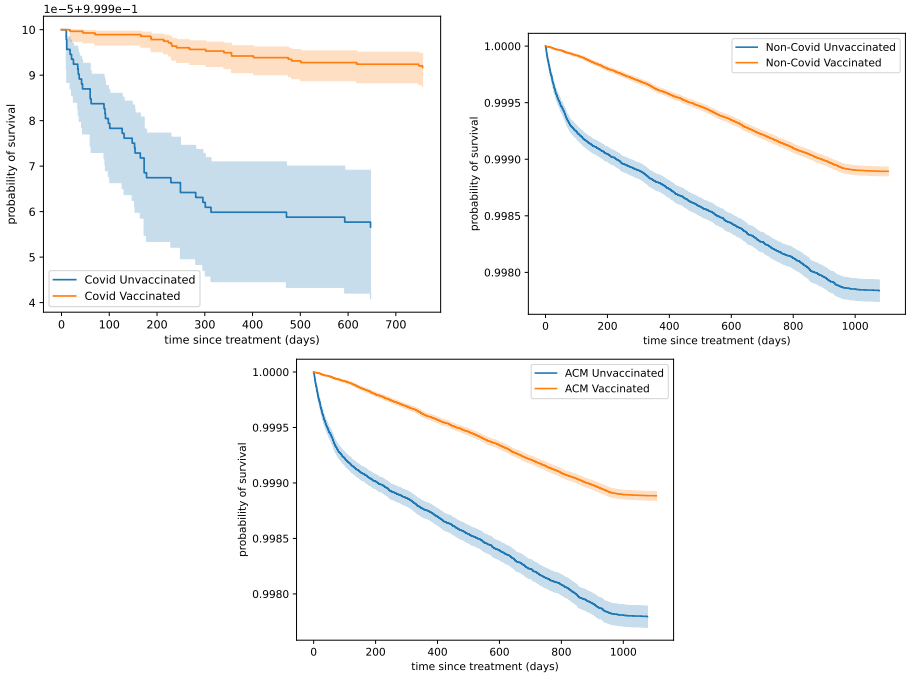



Figure A.8: Survival curves of vaccinated and unvaccinated individuals of age cohort 1980-2000, no WLZ, for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively.

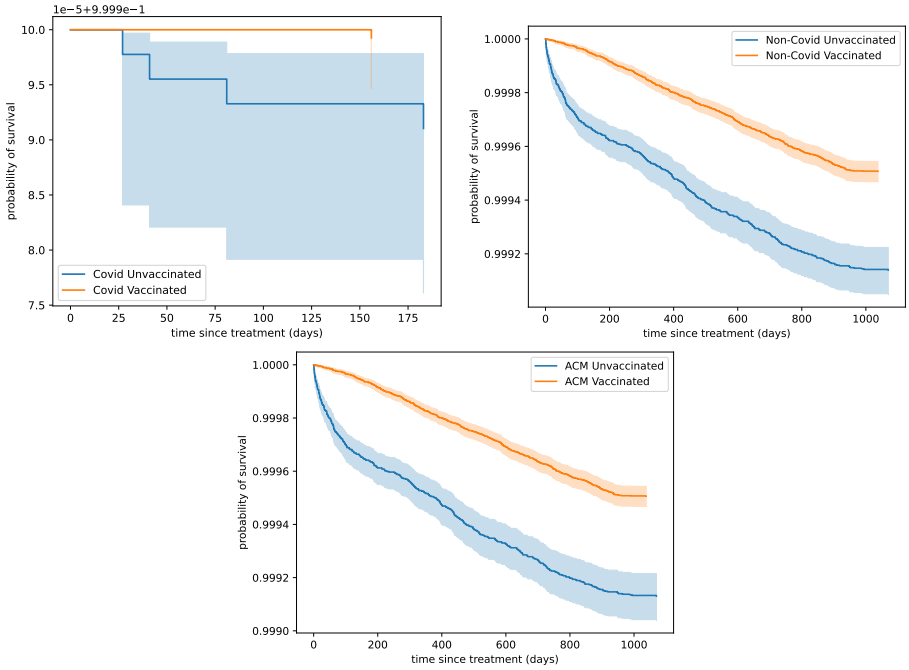



Figure A.9: Survival curves of vaccinated and unvaccinated individuals of age cohort 12000-2020, no WLZ, for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively.

*Appendix A.8. Kaplan-Meier survival analysis data based on the traditional fully vaccinated definition*

In this section we present Kaplan-Meier data for the 1940-1950 age group, the age group we focus most on in this paper, based on the more traditional vaccination status definition of "fully vaccinated" two weeks after completion of the primary course (typically consisting of two doses). This allows for additional comparisons, including comparison with the other Kaplan-Meier survival data plots, where follow-up for the "vaccinated" starts immediately after dose 1. Most people in the Netherlands, and especially most older age groups, received a two-dose primary vaccination series (mostly the Pfizer or Moderna mRNA vaccines). Some people received the two-dose AstraZeneca primary vaccination series; and some people (especially younger age groups) received the one-dose Janssen (Johnson & Johnson) primary vaccination. The adapted matching algorithm can handle all cases. For most people in the Netherlands receiving a two-dose primary vaccination, the time between their primary vaccination dose 1 and dose 2 was between 3 and 6 weeks.

Other than this change of vaccination status definition, we keep using the same fixed cohort, "never-treated" (unvaccinated) vs. "ever-treated" (vaccinated) design, using the same matching on age, sex, and comorbidities, and using the same PTDM (Prescription Time-Distribution Matching) requirement [42]. Thus, when selecting the comparison group with status unvaccinated, the matching algorithm selects never-treated individuals and it requires that an unvaccinated person, who is to be matched with a vaccinated person, is alive on the "fully vaccinated" date of that vaccinated person. Follow-up for each individual starts on that same date.

Similar to before, Figure A.10 shows Kaplan-Meier survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, individuals without WLZ (long-term care), for mortality from Covid; cause of death other than Covid; and ACM, respectively. Figure A.11 shows the same data for individuals *with* WLZ. The results do not change substantially based on this different vaccination status definition/design choice, compared to the results shown in Sections 3.2 and Appendix A.7. This suggests that the results do not strongly depend on this definition/design choice. In other words, the observed results are not strongly dependent (only) on data for vaccinated and unvaccinated individuals in the period immediately after dose 1 until shortly after dose 2.

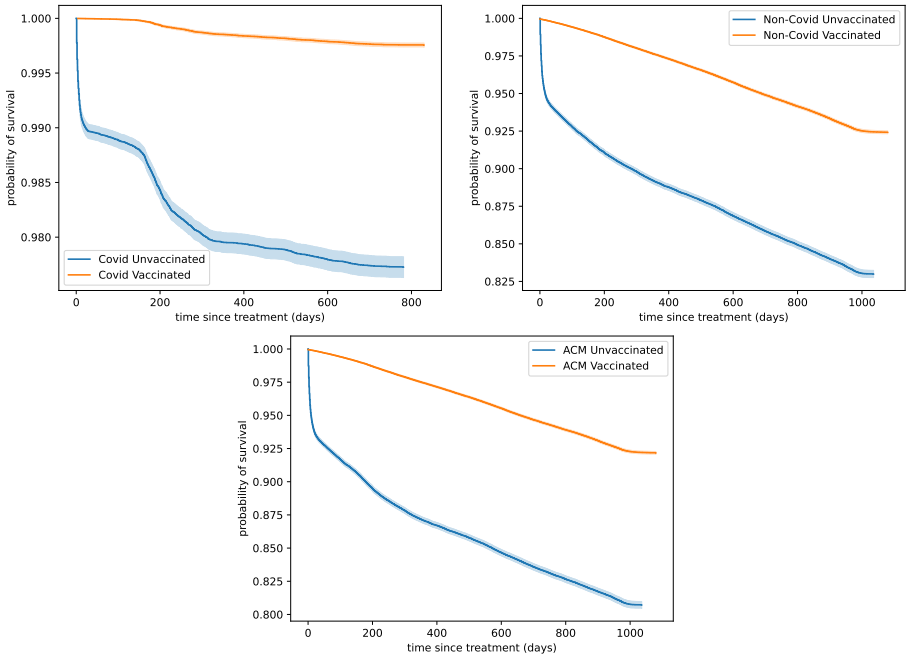



Figure A.10: Survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, no WLZ (long-term care), for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively; now based on the traditional definition of fully vaccinated two weeks after completion of the primary course.

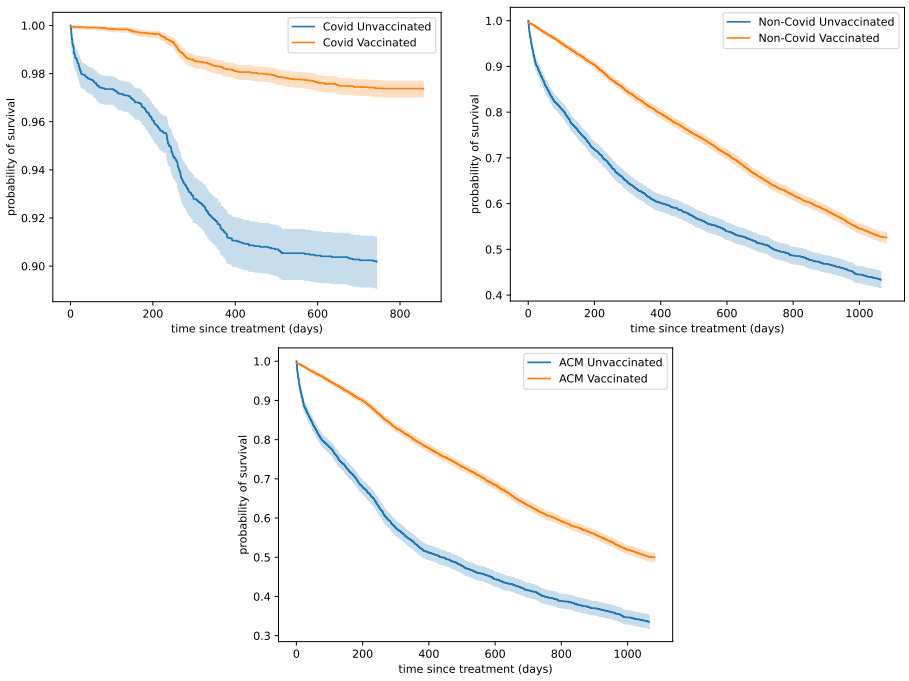



Figure A.11: Survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, *with* WLZ (long-term care), for mortality from Covid (top left); cause of death other than Covid (top right); and All-Cause Mortality (ACM, bottom), respectively; now based on the traditional definition of fully vaccinated two weeks after completion of the primary course.

The results are slightly different in that the sharp dip in survival percentage indicating unusually high mortality for those with status unvaccinated immediately after start of follow-up is slightly less pronounced. For individuals with WLZ (Figure A.11), this difference in KM survival curves is more notable (compare with Figure A.4) than for individuals without WLZ. All this is in line with the observations in all other analyses: many individuals with status unvaccinated died in the period of dose 1 and immediately after; this effect is strongest in the most vulnerable subgroups; but these effects are not limited to dose 1 and immediately after but persist for a considerable amount of time.

*Appendix A.9. Additional calendar time analysis data for various age cohorts*

This section shows calendar time analysis graphs similar to those in section 3.4. Graphs are shown both for relatively old and vulnerable age groups and for younger groups. See the main text for explanation of the graphs.

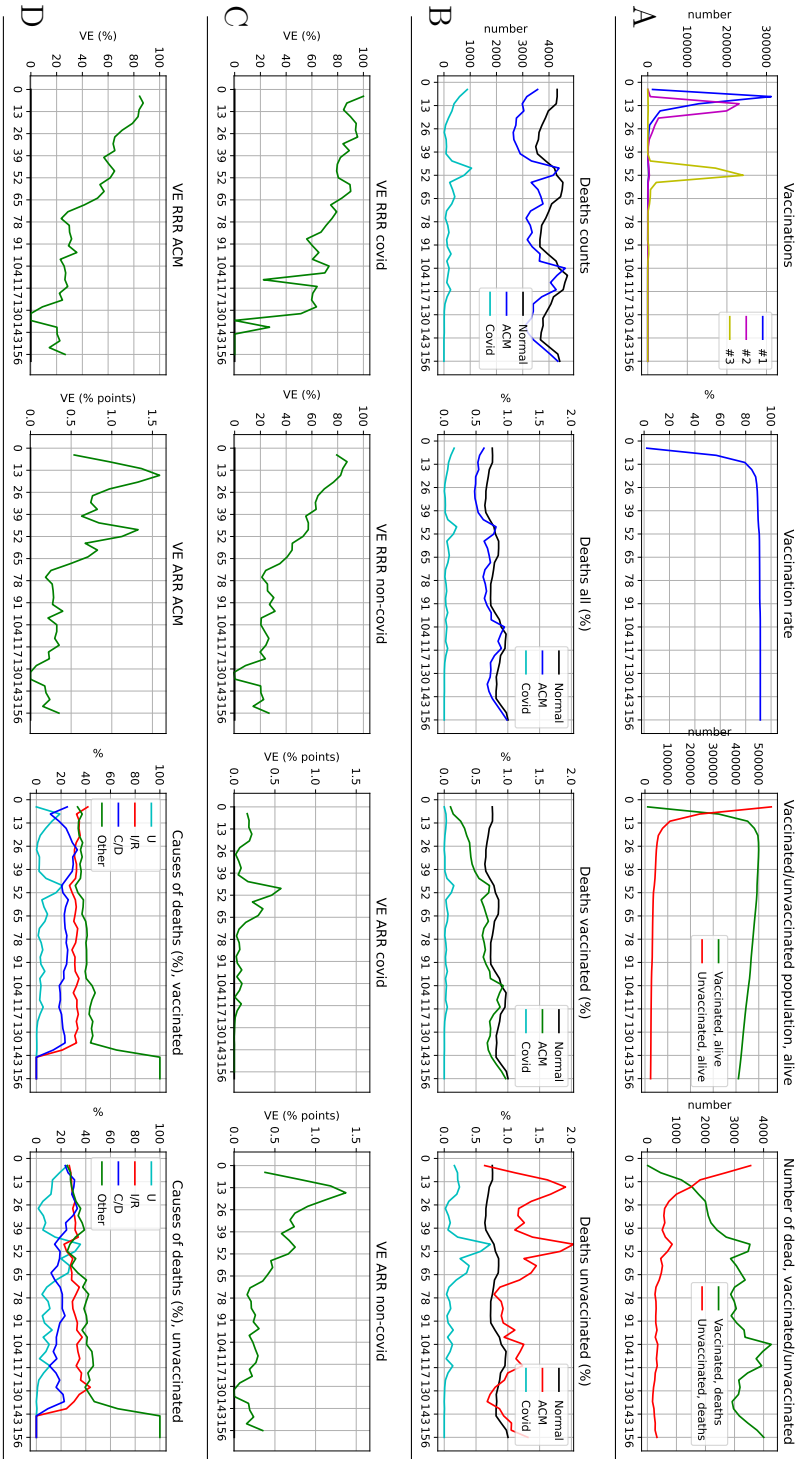



Figure A.12: Per-4-weeks calendar time data of age cohort 1930-1940 (unmatched), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

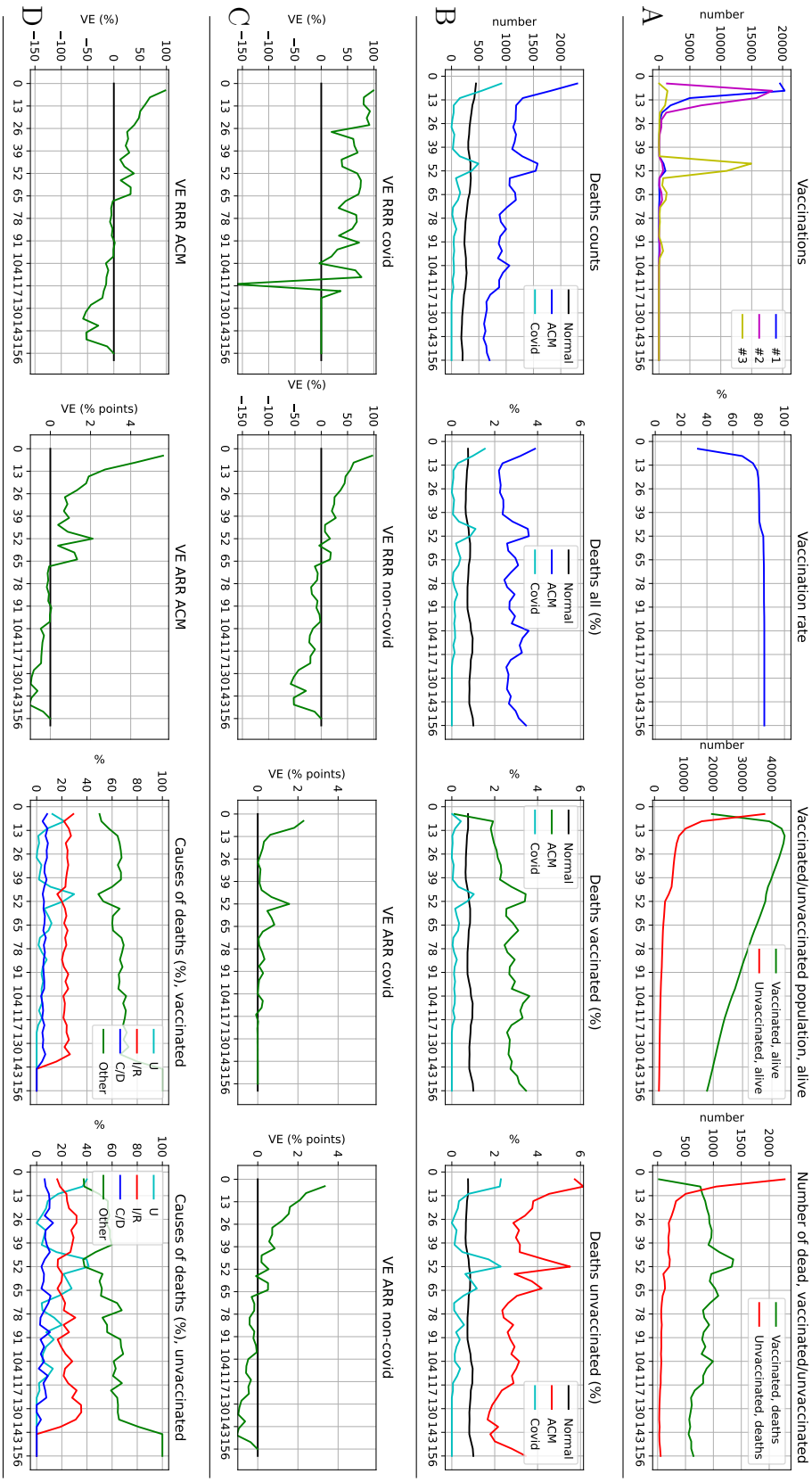



Figure A.13: Per-4-weeks calendar time data of age cohort 1930-1940 (unmatched), with WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

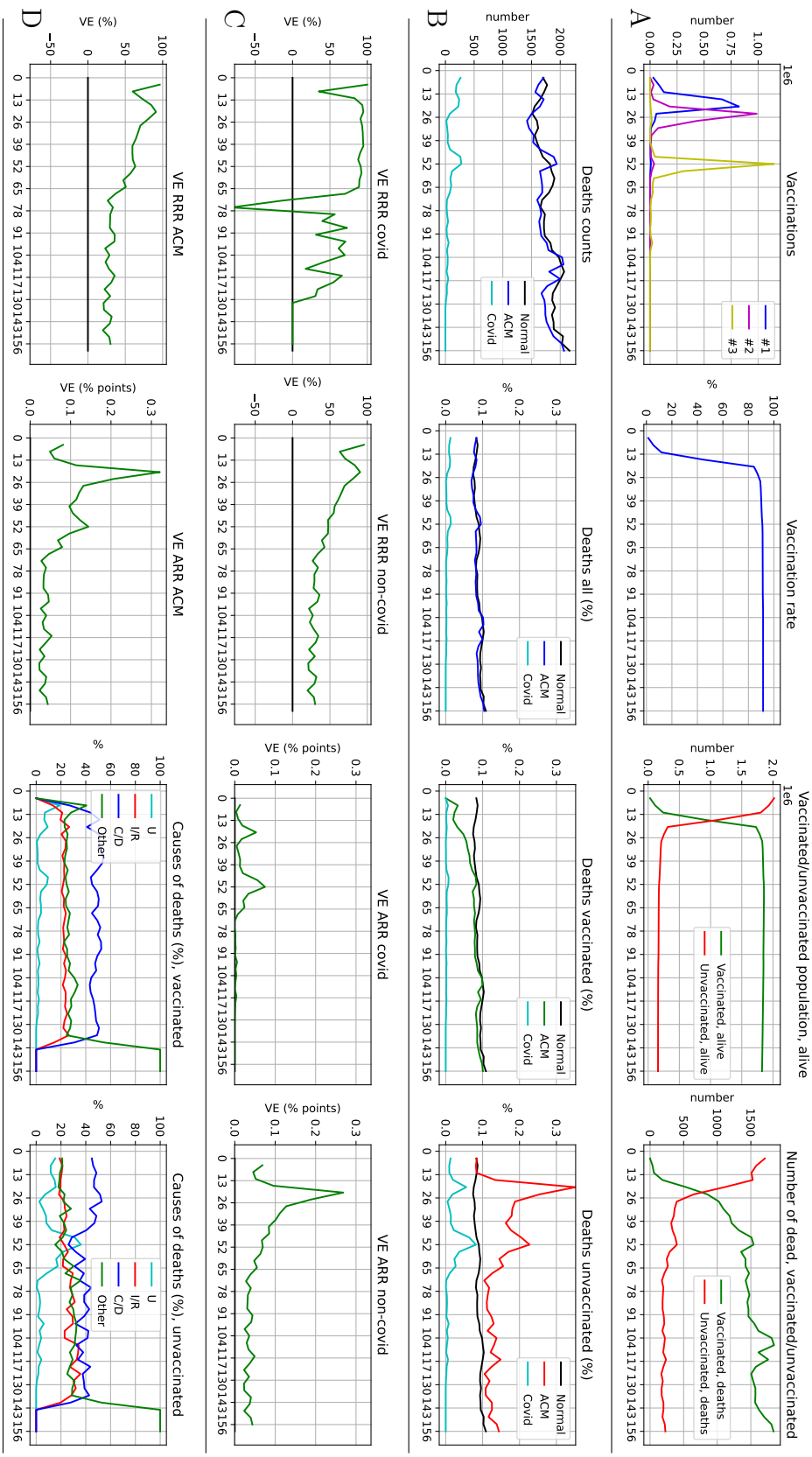



Figure A.14: Per-4-weeks calendar time data of age cohort 1950-1960 (unmatched), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

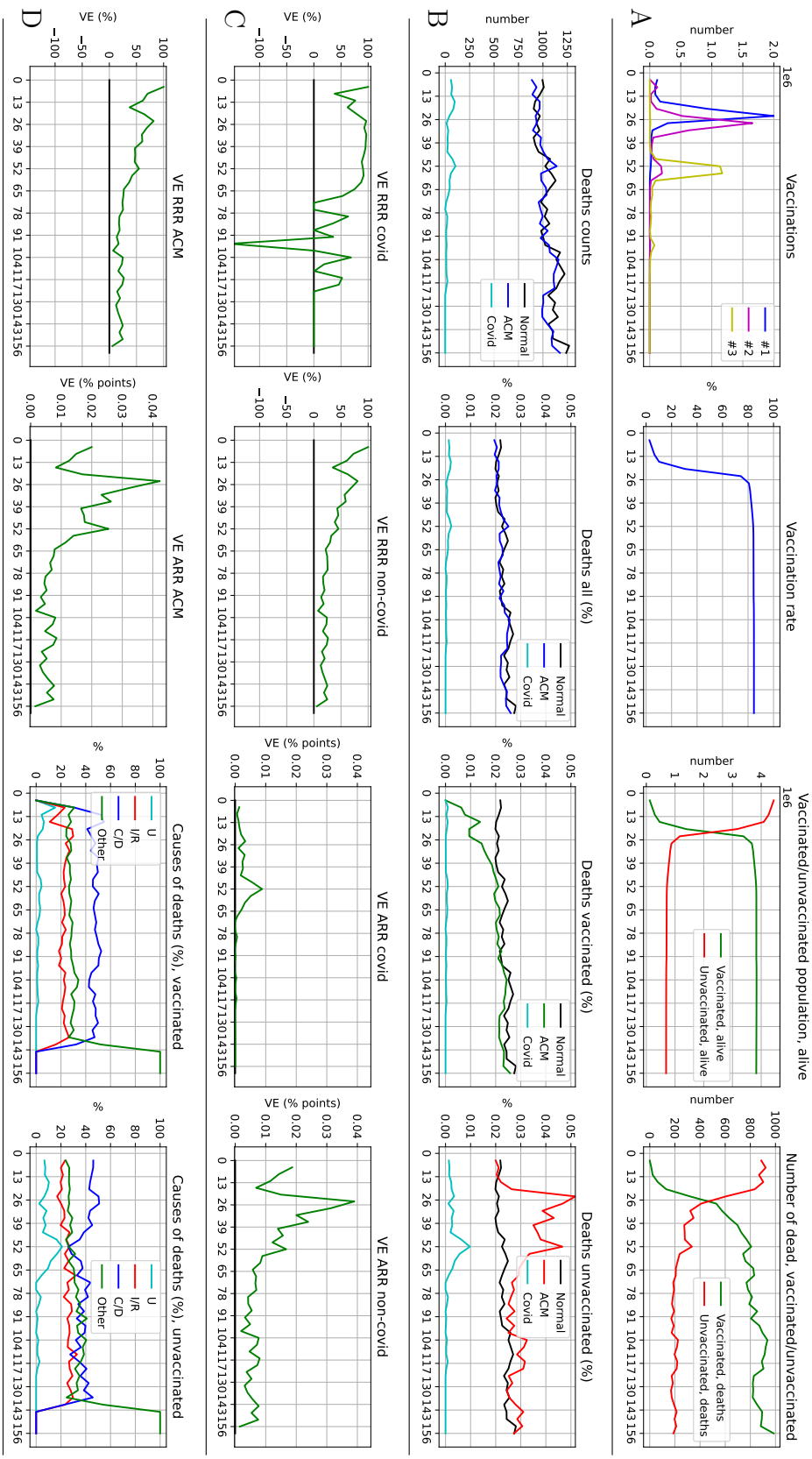



Figure A.15: Per-4-weeks calendar time data of age cohort 1960-1980 (unmatched), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

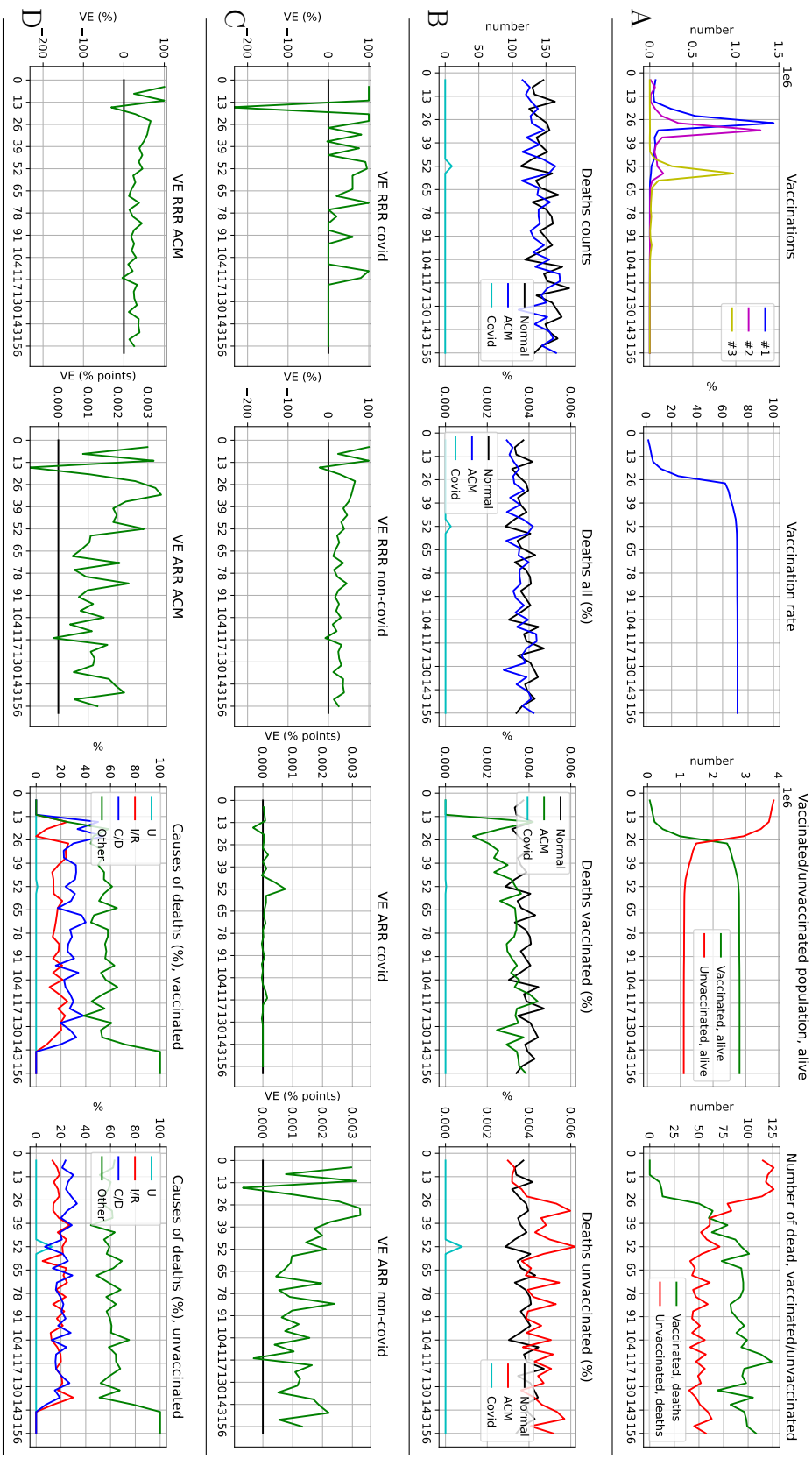



Figure A.16: Per-4-weeks calendar time data of age cohort 1980-2000 (unmatched), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

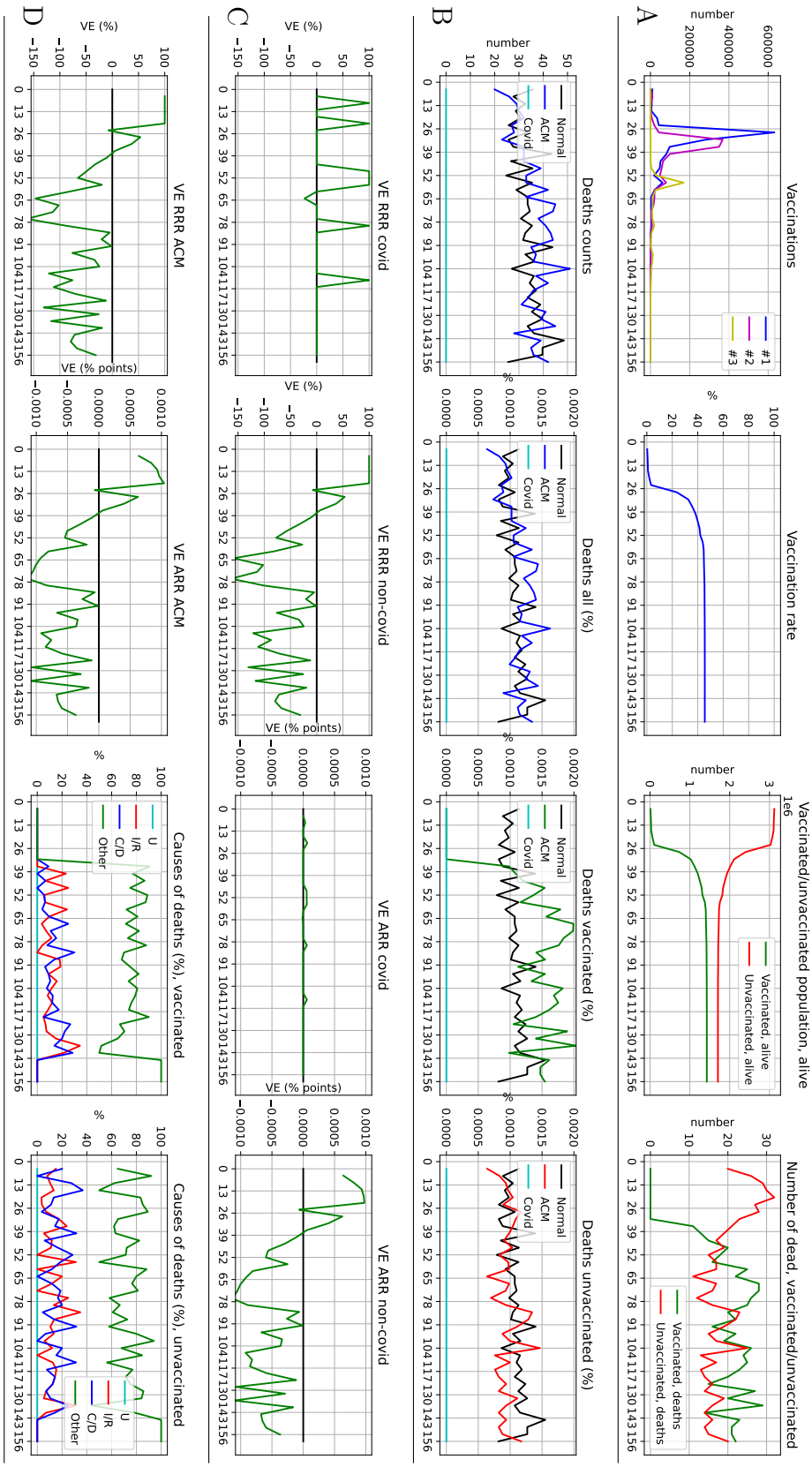



Figure A.17: Per-4-weeks calendar time data of age cohort 2000-2020 (unmatched), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

*Appendix A.10. Calendar time analysis data based on matched data*

This section shows, mainly for illustration purposes, some calendar time analysis graphs similar to those in Section 3.4 and Appendix A.9, but now based on matched data, as explained in Section 2.6. Graphs are shown for the relatively old and vulnerable age group born 1940-1950, both without and with long-term care (WLZ), which are the groups that most analyses in this paper focus on. See the main text for explanation of the graphs.

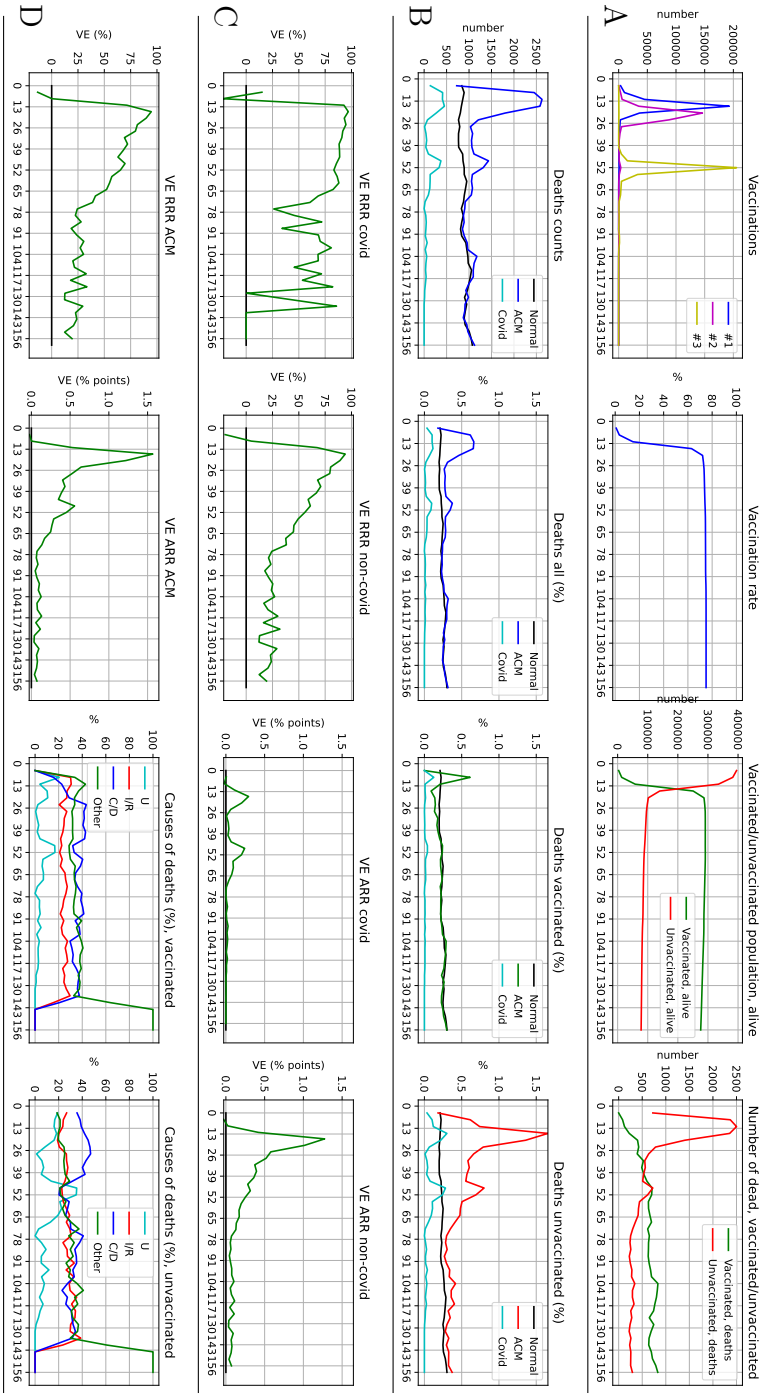



Figure A.18: Per-4-weeks calendar time data of age cohort 1940-1950 (*matched*), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

Compared to the unmatched calendar time data graphs, observed effects in the unvaccinated subgroup are somewhat magnified thanks to the 3:1 vaccinated/unvaccinated ratio enforced in our matching algorithm. Otherwise, the patterns are similar. This reinforces the idea that the differences observed in the unmatched calendar time data analyses are mainly due to *unmeasured* confounding and artifacts; matching based on measured data does not meaningfully change these calendar time-based results.

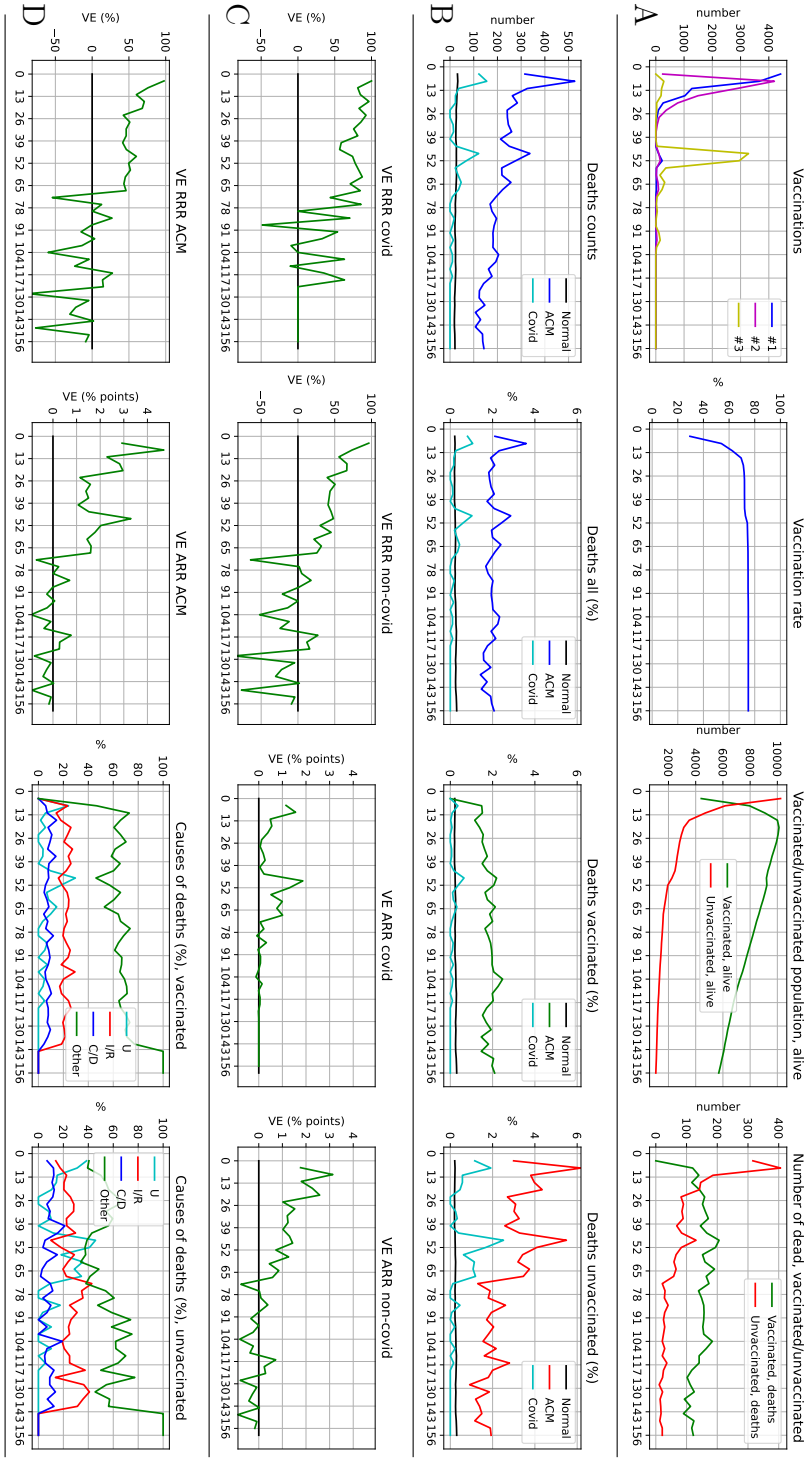



Figure A.19: Per-4-weeks calendar time data of age cohort 1940-1950 (*matched*), with WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

*Appendix A.11. Confounding-by-indication bias*

Our matching and analysis approach (described in detail in Appendix A.5) cannot fully eliminate bias, esp. confounding-by-indication bias [27, 29]. In this section we provide some extra insight into this remaining bias.

Within confounding-by-indication bias, for our purposes it is conceptually useful to explicitly distinguish between confounding-by-indication due to high vulnerability of the individual, and confounding-by-indication due to having died coincidentally just before they would be vaccinated. This relates to "intention-to-treat" considerations in the analysis of both clinical and observational studies [88, 80]. The first case is related to more typical healthy vaccinee effects, and typically means that (at the start of the vaccination round) there was no intent to vaccinate the individual (due to being too vulnerable). The second case instead corresponds to a situation where (at the start of the vaccination round) there *was* an intent to vaccinate the individual; but the vaccination round takes time, each week a number of people are expected to die, and someone who happens to die just before they would otherwise have been vaccinated necessarily falls in the 'unvaccinated' category. Our matching, selection, and start of follow-up calculation algorithm is somewhat sensitive to that second case. This is because an unvaccinated individual alive at the vaccination moment of its vaccinated contemporaries may have 'intended' to be vaccinated in the next few weeks, but coincidentally dies just before that intended vaccination moment. Such unvaccinated individuals may be selected in our randomized matching and selection process and contribute to unvaccinated deaths shortly after start of follow-up. And because, due to the very high vaccination rate for most age groups, the number of unvaccinated individuals in general is relatively small, the impact of this particular subgroup might be substantial.

We note that no analysis method is immune to this particular bias. Nevertheless, it is interesting and important to better assess the seriousness of that particular bias coming from that specific subgroup for our analyses, specifically our KM-based survival analyses. For this reason, in the following we quantitatively estimate its expected impact. (This quantitative model is available to other researchers, see Section 9.) This is possible based on the combination of vaccination data and normal, expected mortality rates we have already collected and calculated for the various age groups. Specifically, we have calculated the likely quantitative impact of this bias for the 1940-1950, no WLZ (no long-term care) group. This is the group that most of our analyses have focused on, and a group that (due to their age) has relatively high mortality per week, so the impact of this specific bias may be relatively high. Figure 5 shows the relevant underlying data on which the calculations are based.

In our estimation model, we use the normal, expected mortality rate per week for this age group as calculated based on data from previous years (see Appendix A.1). For this age group this is (roughly) 0.05% per week. In our simplified model, we use this assumed mortality rate for everyone, regardless of vaccination status and ignoring underlying health differences and the possible impact of Covid-19, so as to 'isolate' the potential impact of this bias we are interested in.

Next, we assume an "intended" vaccination rate of 91.7%. This is also realistic, and derived from the actual data for this age group after vaccination round 1. Based on the calendar time data from the actual first Covid-19 vaccination round in 2021 (see Figure 5), which from start to finish took approx. 36 weeks, our model simulates the vaccination campaign, per week. Each week a number of people are vaccinated, a number of people remain who intend to be vaccinated in the subsequent weeks, and a number of people remain who do not intend to be vaccinated. The number of vaccinations per week has a strong peak, roughly in the middle of that 36-week period (see Figure 5). At the same time, in our model each week a number of people die in each subgroup, based on that expected mortality rate per week. In this way, we can estimate relatively accurately the expected number of people in each subgroup per week – including those who intended to be vaccinated but who happen to die before their intended vaccination. This is exactly that particular subgroup we are interested in.

Next, based on all these numbers, we simulate our matching and selection process (see Appendix A.5). That process randomly matches vaccinated and unvaccinated individuals using a 3:1 ratio. Note that here we ignore the matching aspects based on underlying comorbidities, sex, and more precise age category;

and we also ignore the known vaccination status missing data issues. But we do use realistic numbers (derived from the actual data) of vaccinated (300,000) vs. unvaccinated (100,000) matched individuals resulting from the process (see Appendix A.6). Because the matching algorithm matches people based on the vaccination week of the vaccinated matches, many more people (both vaccinated and unvaccinated) are selected based on the weeks in which many vaccinations took place (roughly in the middle of the 36-week period); and only very few from the very early and from the very late vaccination weeks. However, in the very early vaccination weeks many intended-to-vaccinate people are yet to be vaccinated, and many subsequent vaccination weeks remain during which those intended-to-vaccinate people may die, so these contribute relatively heavily to the specific bias we are analyzing here.

Finally, based on this simulated process, we can calculate the expected impact of this specific, intention-to-treat related bias. The simulated matching and selection process leaves us with a (simulated) group of matched vaccinated and unvaccinated individuals. The unvaccinated group, based on these parameters and this model, consists of unvaccinated individuals who did not intend to be vaccinated (99.21%), together with a small number of unvaccinated individuals (0.79%) who intended to be vaccinated (but died before intended vaccination). For each of these subgroups we calculate the expected cumulative mortality probability for 36 weeks since the start of simulated start of follow-up. For both the vaccinated subgroup and the unvaccinated, don't-intend-to-vaccinate subgroup this is simply the cumulative probability based on the assumed normal mortality probability per week. That mortality probability per week of 0.05% leads to a cumulative mortality probability over 36 weeks of 1.79%. But for that very small unvaccinated, intended-to-vaccinate subgroup the cumulative mortality probability is obviously 100%, since they are *defined* by their having died before intended vaccination, within the 36-week vaccination period. Altogether, for the total unvaccinated group this leads to a total mean cumulative mortality probability of 2.56%; thus, the bias due to this unvaccinated, intended-to-vaccinate but died subgroup corresponds to an expected extra 0.77 percentage points of cumulative mortality.

We can compare these values to the actual, measured cumulative mortality values we have observed for this age group after matching. These correspond the Kaplan-Meier data shown in Figure 3. For the vaccinated group, the actual cumulative mortality after 36 weeks comes very close to the simulated, expected value, and corresponds to approx. 1.6%. This actual, realized value is a bit lower than the calculated expected value of 1.79%; presumably due to the healthy vaccinee effects discussed previously and/or true protective effects of the vaccine. However, for the unvaccinated subgroup in this age group the actual, measured cumulative mortality after 36 weeks is much higher, namely 14.1% (and even after only 4 weeks it is already 8.6%). Thus, in that light the estimated bias of 0.77% is a virtually negligible bias. All this means that the extreme short-term mortality effect we observe in the KM curves for the unvaccinated has little to nothing to do with this particular form of selection bias. Instead, it must have other causes (as described in this paper).

*Appendix A.12. Analysis of the first booster vaccination round*

Previous work on the same data [18, 24] has reported that the (first) booster vaccination round at the end of 2021 in The Netherlands brought back waning VE levels to very high levels. Based on this they conclude that this calculated renewed protection offered by booster vaccination argues against healthy vaccinee effects playing a major role in all the results; and that it reaffirms that the vaccinations really do offer biomedically potent, but perhaps somewhat short-lived, protection. To assess this argument, we similarly do an additional analysis of the first round of booster vaccinations in late 2021. This is an analysis similar to the calendar time analyses of section of 3.4, but smaller in scope.

For this analysis, we look again at the 1940-1950 age cohort analyzed extensively before. Figure A.20 and Figure A.21 show the same type of per-4-weeks calendar time data as in the previous calendar time analyses, for this 1940-1950 age cohort. These are again based on unmatched data, for the group without and with WLZ (long-term care), respectively. Importantly, in these data and graphs, for the purposes of this analysis, people are considered "vaccinated" if and only if they have received the late 2021 booster vaccination (administered since November 2021). For the vast majority of people, this corresponds to dose 3. Being "unvaccinated", in this analysis, means that a person has either received no vaccinations at all (including doses 1 and/or 2), or – as is true for the vast majority – has received the primary vaccination series but no booster.

About 1.3 million people out of a total of about 1.5 million people in the 1940-1950 age cohort (without WLZ) had received at least 1 vaccination dose by the first half of 2021 (see Figure 5). Thus, only about 200,000 people had no (registered) primary series vaccination. Figure A.20 shows that about 1.25 million people, from this age cohort, received another (usually third) vaccination, in late 2021 or early 2022; and thus is considered "vaccinated" in this booster analysis.

In terms of mortality and VE patterns we see again essentially the same patterns as what we saw for the first (primary series) vaccination. At the beginning of the booster campaign, we see a brief mortality peak among vaccinated (now: boostered) individuals. This may, as during the primary series vaccination campaign, be related to the policy of providing booster vaccinations first to the most vulnerable people, who have higher mortality risk anyway (i.e. transient negative HVE). But it may also have other causes, related to recent vaccination together with concurrent SARS-CoV2 infection, as Covid mortality has also increased among this group. (This period coincides with the so-called Delta variant wave of infections, which later transitions into subsequent Omicron waves.)

Subsequently, we again see the familiar pattern of very high apparent 'protection' by the vaccine, both for covid and for all other causes of death, directly related to extremely and unrealistically high mortality among the 'unvaccinated' (in this case: those who have not received a booster vaccination in this round). That subgroup has a mortality level that is much higher than what is normal for the age cohort. According to the causes of death (row D) this can be (only) partially attributed to Covid-19. Other and more frequent causes of death are the usual causes of death: cancer (C/D), cardiovascular disease (I/R), and "other" (which includes, for example, dementia and a variety of other neurological disorders associated with old age).

Finally, as before we see markedly increased *overall* mortality as well (row B); that is, increased mortality of boostered and unboostered people together. The peak of unvaccinated (unboostered) and overall mortality coincides, as in the primary series vaccination round, with the peak of this booster vaccination round, but also with a (limited) peak of Covid mortality. As discussed in detail in previous sections, all this again points to large healthy vaccinee effects playing a role, possibly in combination with vaccine status misclassification bias. There is an unclear role for concurrent SARS-CoV2 infection, both for unvaccinated (unboostered) and (recently) vaccinated people.

Among WLZ recipients in the 1940-1950 age cohort (Figure A.21), we see essentially the same picture. The main difference is that for this subgroup the booster vaccination rate stagnates at a value between 50 and 60%. This may be because many in this group were already too vulnerable to receive another Covid vaccination, which would be in line with the HVE hypothesis. It may also be the case that, as

hypothesized for the primary series vaccinations, relatively many vaccinations remain unregistered, in line with the misclassification bias hypothesis.

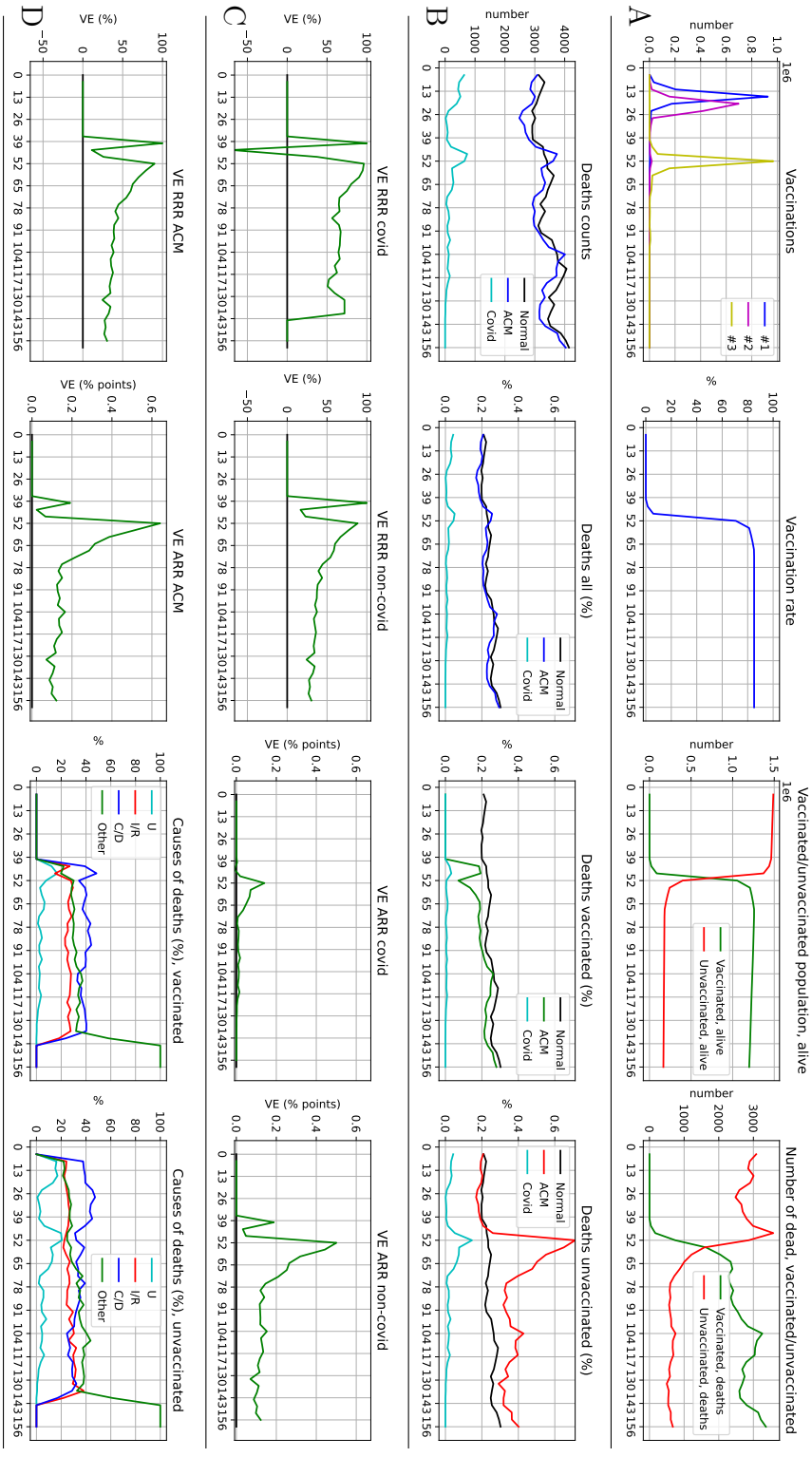



Figure A.20: Per-4-weeks calendar time data of age cohort 1940-1950 (unmatched), no WLZ, 2021-2023, booster analysis. Here, the definition of having been 'vaccinated' is based on having received a booster dose at the end of 2021. See text for detailed explanation.

In short, this analysis suggests that the same data and methodological issues and artifacts play a large role in the booster round(s) as in the primary series vaccinations. And it suggests that for the same reasons VE calculation are detrimentally affected, and apparent renewed high booster vaccine protection [18, 24] is artificially inflated by these artifacts.

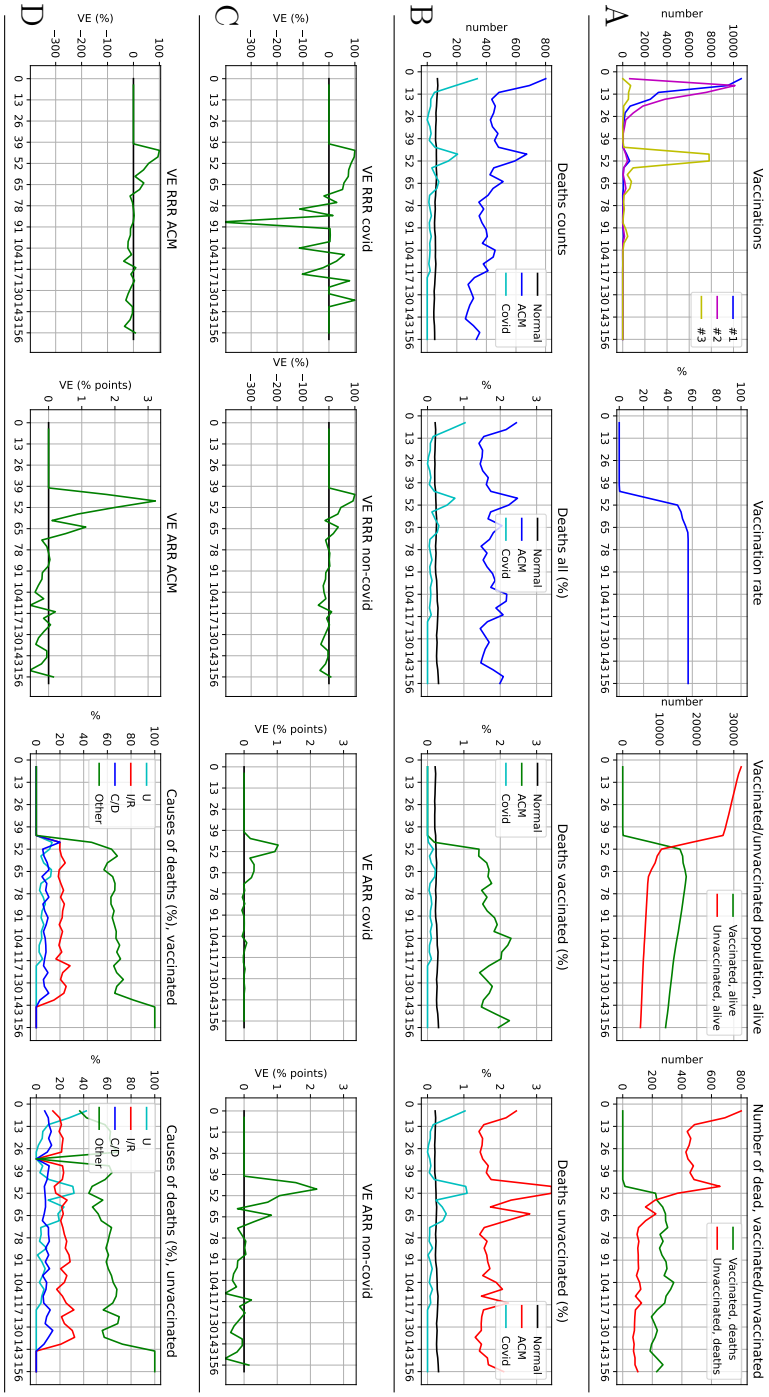



Figure A.21: Per-4-weeks calendar time data of age cohort 1940-1950 (unmatched), with WLZ (long-term care), 2021-2023, booster analysis. Here, the definition of having been 'vaccinated' is based on having received a booster dose at the end of 2021. See text for detailed explanation.

*Appendix A.13. Additional statistics on vaccination data completeness and healthy vaccinee bias*

In this study, we have found multiple indications of both residual HVE-related confounding and vaccine status misclassification bias playing a role in the effects. Here we collect and calculate various additional statistics and other information relevant to those hypotheses; both based on our own data sources and on external data sources.

Table A.3 contains calculated vaccination rates for "low-risk" and "high-risk" individuals, respectively, per age and WLZ group, based on the CIMS dataset, with "low-risk" and "high-risk" defined again on already calculated Rx-Risk scores [45] derived from previous year medication use, and used in this study for differentiating and matching based on estimated comorbidity. Low-risk here corresponds to a weighted Rx-Risk score of 15 or lower; high-risk corresponds to values above that (see Figure A.2 and A.3 for Rx-Risk distributions for the 1940-1950 age group).

We observe in the table that with the exception of the youngest age group (where many healthy young people chose to remain unvaccinated), the calculated vaccination rate is markedly lower for high-risk individuals individuals than for low-risk individuals. The difference in calculated vaccination rates is largest for the oldest and most vulnerable groups and amounts to approx. 5 to 7 percentage points. On the one hand, this is compatible with the residual HVE hypothesis, as some extremely vulnerable and close to death individuals were likely not vaccinated. At the same time, the large size of this difference also supports the hypothesis of significant vaccine status misclassification bias. Health policy at the time recommended Covid-19 vaccination for most people including (and especially) vulnerable people with only few exceptions [24]; thus, it seems unlikely that this many high-risk individuals remained unvaccinated. In other words, this suggests for those high-risk individuals, relatively many vaccinations were not registered in CIMS.

Table A.3: Calculated vaccination rates for low-risk and high-risk individuals.

| age cohort | WLZ | nr. people low risk | % vacc. | nr. people high risk | % vacc. | % at high risk | diff. vacc. % low–high risk |
|---|---|---|---|---|---|---|---|
| 1930-1940 | 0 | 548757 | 91.76 | 21856 | 85.59 | 3.830 | 6.17 |
| 1940-1950 | 0 | 1463402 | 93.19 | 28417 | 88.3 | 1.905 | 4.89 |
| 1950-1960 | 0 | 2025306 | 91.56 | 24724 | 87.81 | 1.206 | 3.75 |
| 1960-1980 | 0 | 4525349 | 84.63 | 31811 | 81.87 | 0.698 | 2.76 |
| 1980-2000 | 0 | 3911189 | 71.74 | 7140 | 68.82 | 0.182 | 2.92 |
| 2000-2020 | 0 | 3131693 | 45.54 | 225 | 67.56 | 0.007 | -22.02 |
| 1930-1940 | 1 | 53848 | 85.04 | 5451 | 78.08 | 9.192 | 6.96 |
| 1940-1950 | 1 | 30311 | 87.13 | 2492 | 81.26 | 7.597 | 5.87 |
| 1950-1960 | 1 | 21660 | 90.49 | 1012 | 83.4 | 4.464 | 7.09 |
| 1960-1980 | 1 | 35892 | 91.45 | 829 | 84.8 | 2.258 | 6.65 |
| 1980-2000 | 1 | 44184 | 82.39 | 317 | 81.07 | 0.712 | 1.32 |

A recent study [19] by the Netherlands Institute for Health Services Research NIVEL, using independent and relatively detailed general practitioner health records of about 1 million Dutch adults (i.e. a substantial subset of all Dutch citizens), also provides strong additional support for both hypotheses. In that study, they have and provide information on the percentage of people receiving a type and level of care suggestive of high vulnerability or even end of life conditions, so-called "high-intensity care" (which is a type of data not available in our study). They measure and report this type of care especially in the four weeks before the actually realized vaccination date (for vaccinated people), or the expected vaccination date (for unvaccinated people, given their age group). Such "high-intensity care" prevalence among all unvaccinated people was significantly higher than among all vaccinated people in their dataset:

0.2% vs. 0.1%, i.e. double the rate. Among the subset who died within 12 months from their actual or expected vaccination date, both the absolute prevalence percentages of this, and the relative differences, were even higher. Approx. 11.7% of those unvaccinated dead individuals received that type of care in the four weeks before their expected vaccination date, as opposed to approx. 3.4% for vaccinated individuals; i.e. approx. 3.5 times higher rate for unvaccinated people. This clearly supports the hypothesis of significant unmeasured HVE-related confounding in our study, related to that "high-intensity care" data which was available in the NIVEL study.

With respect to the vaccine status misclassification issue, that same NIVEL study also has relevant data, as their dataset contains vaccine records independent of the central national CIMS vaccination registry, for their cohort. Their analysis confirms that many 'unvaccinated' people according to CIMS were actually vaccinated. Based on their data, they were able to reduce the percentage of vaccinated but unregistered people from an estimated 7% to 4.5% (a relative 35% reduction). In all, they estimate that 25% of those labeled 'unvaccinated' likely received at least one vaccination, and this could be as high as 75% in the oldest age groups [19].

Table A.4: Estimated numbers of vaccinated registered, vaccinated unregistered, and truly unvaccinated individuals in the different age/WLZ cohorts, and estimated percentage vaccinated unregistered among those with status "unvaccinated" (last column).

| age cohort | WLZ | vaccinated registered | vaccinated unregistered | truly unvaccinated | % vaccinated among status "unvaccinated" |
|---|---|---|---|---|---|
| 1930-1940 | 0 | 522246 | 40831 | 7536 | 84.4 |
| 1940-1950 | 0 | 1388836 | 88065 | 14918 | 85.5 |
| 1950-1960 | 0 | 1876080 | 142343 | 31607 | 81.8 |
| 1960-1980 | 0 | 3855847 | 292861 | 408452 | 41.7 |
| 1980-2000 | 0 | 2810801 | 212063 | 895465 | 19.1 |
| 2000-2020 | 0 | 1426325 | 107373 | 1598220 | 6.3 |
| 1930-1940 | 1 | 50048 | 7906 | 1345 | 85.4 |
| 1940-1950 | 1 | 28435 | 4040 | 328 | 92.5 |
| 1950-1960 | 1 | 20444 | 2001 | 227 | 89.8 |
| 1960-1980 | 1 | 33526 | 2828 | 367 | 88.5 |
| 1980-2000 | 1 | 36660 | 5497 | 2344 | 70.1 |

Importantly, the misclassification issue has a relatively large potential impact in particular because in most age cohorts in The Netherlands, the vaccination rate is very high, such that the number of actually unvaccinated people is very low. As a simple arithmetic example, in a cohort of 1 million people where 93% of all people are vaccinated, only 70,000 people are unvaccinated. If, at the same time, 7% of the 930,000 vaccinated people remain unregistered, that amounts to 65,100 people, which then corresponds to 48% of all people labeled as "unvaccinated" (i.e., no vaccine registration), with only 52% being truly unvaccinated. And in reality, probably a higher percentage of vaccinations is missing, strengthening the effect. Furthermore and as shown above, among those truly unvaccinated individuals a substantial percentage of people appear to be unvaccinated due to extreme vulnerability; leaving even fewer 'normal, relatively healthy' unvaccinated people in the control group. At the very least, this confirms suspicions of large heterogeneity in the research groups (see Section 2.5).

Based on the same logic, Table A.4 shows estimates of those percentages "vaccinated but unregistered" among those with status "unvaccinated" (last column), derived from our study's data and in particular the data in Table A.3. The calculations are done separately for low-risk subgroups and high-risk subgroups (as we know that missing vaccination registration percentages are likely different for those groups), and subsequently combined. The calculations are based on estimated parameter values (see the other data

and estimates in this section) assuming that for age cohorts without WLZ, for low-risk individuals the probability of a missing vaccine registration is 7% and for high-risk individuals that probability is 15%. For age cohorts with WLZ, for low-risk individuals the probability of a missing vaccine registration is assumed to be 13% and for high-risk individuals that probability is 20%. These are approximate estimates. Given these parameters and data, for older and vulnerable age groups the percentage "vaccinated but unregistered" among those with status "unvaccinated" is around 85% or even more (thus, even higher than the estimates from NIVEL described above). This percentage is only relatively low for younger (and healthier) age groups, where both overall vaccination rate is lower *and* there are relatively few high-vulnerability individuals, who have higher likelihood of missing vaccine registration.

Other studies confirm significant gaps in the CIMS vaccination, and indicate significant errors and gaps unrelated to lack of patient consent. A cohort study aiming to follow a cohort of vaccinated individuals in The Netherlands [31] found that the central CIMS vaccination registry matched self-reported vaccinations completely only in about 50% of cases. Many self-reported vaccinations were missing in CIMS or with very different dates. There were also a substantial number of cases (3.7%) where the individual claimed to have given consent for CIMS registration but the individual could not be found in CIMS at all. Thus, missing vaccination data does not result from lack of individual consent alone, and there appear to many gaps or errors. In line with all this, both early (end of April 2021) and later (2022) reports from the national Covid crisis management committee (OMT) and others in the Netherlands acknowledge and express concern about large numbers of CIMS vaccination registrations missing during vaccination round 1, in particular for those vaccine administration agencies responsible for vulnerable groups [24, 37, 57]. For those agencies responsible for vulnerable groups, those reports indicate that the percentage of missing data is likely significantly higher than the 7% estimate reported earlier [18], and at the end of April 2021 was around 30%. It is unclear to what extent those early missing vaccinations were later recovered and registered.

Our finding of relatively low calculated vaccination rate in persons receiving long-term care (WLZ) after vaccination round 1 (see Section 3.4), in contrast with what was previously reported [37] and their priority in vaccination schedules, provides additional support for this missing data hypothesis. In line with this finding, during the booster round in the fall/winter of 2021 relatively many elderly recipients of long-term care appear to receive their first vaccination. As vaccination rates barely increase in other groups, it seems more likely that the primary vaccination series was simply not properly and completely registered for these persons.

Table A.5 shows estimated vaccination completeness, by vaccine administration agency. These are estimates based first of all on estimated (rough) total number of vaccinations (second column) per agency (first column), as provided by official government research institute reports [37]. In the vaccine administration agencies column, 1 corresponds to GGD (central vaccination centers, i.e. mostly for relatively healthy individuals capable of going there using personal transport); 2 corresponds to General Practitioners (GP); and 3 corresponds to "Other", corresponding mostly to hospitals, care homes, and similar. The latter category has some, but only some, overlap with the WLZ long-term care indicator derived from insurance data, used in the cohort stratifications and many of the analyses in our study. These reports [37] indicate that those total numbers estimates are based on a combination of CIMS data *and*, because of CIMS gaps, estimates derived from vaccines shipped to the various agencies (in particular for vaccination agencies 2 and 3). The third column of the table contains the number of registered and non-anonymous vaccination round 1 vaccinations in our research CIMS dataset, a version from and updated up until August 2023. That latter dataset also contains, for each vaccination, the same corresponding code for the three types of vaccine administrator. Although these estimates are far from precise, they indicate substantially lower coverage percentage for the agencies associated with the more vulnerable groups, and in any case far more missing data than the 7% missing which is typically mentioned [18, 19].

Finally, there are other interesting and relevant data points related to studies and datasets on reported Adverse Events (AEs) following Covid vaccinations. In the Netherlands, data on such AEs were originally collected by Dutch pharmacovigilance organization Lareb, and subsequently also provided to the

Table A.5: Estimated vaccination registration completeness for vaccinations in vaccination round 1, by vaccine administration agency.

| admin. | est. total vacc. | total CIMS vacc. | % registered |
|---|---|---|---|
| 1: GGD | 20975190 | 18811609 | 89.7 |
| 2: GP | 2315146 | 1902356 | 82.2 |
| 3: Other | 1423326 | 1135648 | 79.7 |

European Medicines Agency EMA. For vaccine batch safety purposes, an important aspect of pharmacovigilance, Lareb [59] used access to the central vaccination registry CIMS to look up batch numbers for those individuals for whom adverse events were reported and batch number was not provided in the AE report itself. For this they used the individual's social security number (translated to the pseudonymized person ID in the CBS Microdata environment) and the vaccination date (with some margin around the date to account for small errors, and other efforts to improve robustness). They (only) did this for sufficiently grave AEs (i.e., excluding very common and mild complaints like "temporary fatigue"). Lareb authors write [59] "Batch numbers were added from CIMS in 73.1% of the requests". This suggests that they were unable to retrieve the batch number from CIMS in 26.9% of cases. Importantly, in CIMS a valid (known) batch number is present for 99.6% of all registered vaccinations. This, together with their generic descriptions of not always being able to retrieve accurate CIMS data, suggests that these vaccinations were not registered at all. Other, independent researchers have worked with presumably the same or very similar but anonymized Dutch AE data, obtained from EMA [60, 61], who (as mentioned) get their data from Lareb. In line with Lareb's own reporting, they report [61] that for Pfizer Covid vaccinations administered in the Netherlands, batch number is missing in a large proportion of reported AE cases. For "non-serious" AEs, in 25.1% of cases batch number was missing; for "serious" AEs, in 34.6% of cases batch number was missing; and for AE reports where "death" was reported, in 36.3% of cases batch number was missing.

Taken together, these various data sources and analyses strongly suggest substantial CIMS vaccine status gaps, in particular for very vulnerable people, and perhaps most of all for deaths, which can lead to severe vaccine status misclassification bias.

*Appendix A.14. Quantitative bias analysis 1: Simulating the mechanisms driving the bias effects*

We use post-hoc quantitative bias analysis (QBA) to better understand and quantify the suspected systematic biases: healthy vaccinee bias and vaccination status misclassification bias. QBA can be understood as a form of sensitivity analysis particularly called for and well-suited in cases where the evidence suggests large residual systematic (non-random) errors [48, 49], as is the case in this study. This includes less "traditional" and less common issues (relevant here) that are related to handling of missing data and exposure/treatment misclassification, which cannot easily be analyzed in other ways. We perform QBA in two different but complementary ways. Both of them are imperfect and mostly exploratory, intended to complement the primary analyses and each other. Although grounded in empirical data, as always with QBA they are dependent on and sensitive to model choices and assumptions.

The first approach, described in this Section (Section Appendix A.15 describes the second approach) focuses on and models the potential underlying mechanisms driving the biases, similar to many other QBA efforts [48, 49]. It uses plausible, mathematically modeled mechanisms and plausible parameters (derived from our study's data) to simulate the biases probabilistically, in an effort to simulate and thereby understand the combined processes that could lead to the observed Kaplan-Meier survival data.

The process starts with a simulated population of (truly) unvaccinated and vaccinated individuals at start of follow-up, as would be expected from our stratification and matching process, for a specific age (and WLZ) cohort. It then simulates what would be expected to happen over time, per day, from start of follow-up onward, in terms of expected mortality for groups with real and apparent vaccination status "vaccinated" and "unvaccinated". The latter includes actually vaccinated but unregistered individuals; modeled through a parameter defining probability of being unregistered despite having been vaccinated, and thus eventually ending up in the category with status "unvaccinated". Thus, the method calculates simulated mortality data over time, given modeled assumptions and parameters representing probability of death per day (per subgroup), probability of no vaccine registration, and also parameters representing underlying and varying vulnerability subgroups. Table A.6 shows the parameters, and example values.

Three vulnerability subgroups are distinguished (modeled): low vulnerability (very healthy), medium vulnerability, high vulnerability. This subdivision into three discrete subgroups is fairly arbitrary; but at least three subgroups (or equivalent highly asymmetric mortality likelihoods) seemed necessary to properly model mortality patterns and differences. Each subgroup comes with its own modeled probability of death per day. The vaccinated and unvaccinated groups can have (and typically do have) different relative proportions of these different vulnerability subgroups, thus modeling potential healthy vaccinee bias. In the parameter settings, the probability of missing vaccination registration can also vary per vulnerability subgroup. This allows modeling of the suspected *conflated* bias effects: high vulnerability individuals are more likely to be truly unvaccinated than medium or low vulnerability individuals within the same (old) age cohort; but also, when the high vulnerability individual is nevertheless vaccinated, they are more likely to have a missing vaccine registration. This is in line with what our own empirical data suggests, as well as external data and reports (see main text). Again in line with what appears to be the situation in the Netherlands vaccination databases, we do not have the possibility of the opposite situation: someone being registered as vaccinated despite not having been vaccinated; this does not appear to have occurred (or only extremely rarely). Thus, the vaccination status misclassification issue is asymmetrical, and is modeled as such.

Additional parameters allow modeling of the suspected temporary, short-term effect of higher (or potentially lower) probability of missing vaccine registration at the start and peak of the vaccination round compared to later, and higher probability of missing vaccine registration in case of death shortly after vaccination. This too is in line with what various data sources suggest may have happened in the Netherlands (see Sections 3.5 and Appendix A.13).

Finally, there are parameters representing variable assumptions about higher probability of death due to Covid-19 for unvaccinated people; and, conversely, potentially somewhat elevated probability of death shortly after vaccination. These parameters model the opposite beneficiary and potentially harmful effects of vaccination, in combination with (simplified) modeled SARS-CoV2 infection wave effects. These effects

Table A.6: Quantitative Bias Analysis 1, example simulation parameters. These parameters are for the age cohort that is the main focus of this paper, the group born 1940-1950 (no WLZ long-term care). This particular set of parameters represents a "mixed scenario" simulation (see text for explanation).

| | | **Vaccinated** | **Unvaccinated** |
|---|---|---|---|
| **Number of people** | N (%) | 300000 (75) | 100000 (25) |
| **Vulnerability group** | N (% in V/U group) | | |
| Low vulnerability | | 280000 (93.33) | 83600 (83.60) |
| Med. vulnerability | | 19600 (6.53) | 8000 (8.00) |
| High vulnerability | | 400 (0.13) | 8400 (8.40) |
| **Vulnerability group** | N (% in whole group) | | |
| Low vulnerability | | 280000 (70.0) | 83600 (20.9) |
| Med. vulnerability | | 19600 (4.9) | 8000 (2.0) |
| High vulnerability | | 400 (0.1) | 8400 (2.1) |
| **Vulnerability group** | probability death per day | | |
| Low vulnerability | | 0.00003 | 0.00003 |
| Med. vulnerability | | 0.002 | 0.002 |
| High vulnerability | | 0.1 | 0.1 |
| **Vulnerability group** | probability missing vacc. reg. | | |
| Low vulnerability | | 0.05 | n.a. |
| Med. vulnerability | | 0.50 | n.a. |
| High vulnerability | | 0.70 | n.a. |
| | | | |
| **Short-term special period** | duration in days | 28 | 28 |
| **Vuln. group, short-term:** | after death: extra prob. missing vacc. reg. | | |
| Low vulnerability | | 0.90 | n.a. |
| Med. vulnerability | | 0.70 | n.a. |
| High vulnerability | | 0.70 | n.a. |
| **Vuln. group, short-term:** | extra prob. death per day | | |
| Low vulnerability | | 0.00015 | 0.000015 |
| Med. vulnerability | | 0.0005 | 0.000005 |
| High vulnerability | | 0.1 | 0.001 |

also come with additional parameters that control a temporary, time-varying effect: we assume, based on observed data, that these effects may have occurred more strongly during the peak of the vaccination round and shortly after, which coincided with a small SARS-CoV2 infection wave.

Based on this combined, fairly complex and sophisticated mechanistic and probabilistic model, the simulation calculates expected mean mortality rates per day for the groups with (apparent) status "vaccinated" and "unvaccinated", under the assumptions defined by the parameter settings. Thus, the simulated group with status "vaccinated" contains only those (simulated) individuals and deaths where their vaccinations were properly registered. The simulated group with status "unvaccinated" contains both truly (simulated) unvaccinated individuals *and* vaccinated but unregistered cases. From the simulated mortality patterns, in turn, we derive simulated Kaplan-Meier survival curves. These are compared to the actual KM curves obtained in our primary empirical data analysis, to assess realism and plausibility.

In essence, this simulation approach models both normal (age stratified) population mortality patterns over time together with the bias factors that, based on the data and analyses of this paper, we expect to play a role. By exploring various parameter settings, we attempt to assess how large each factor needs to be to somewhat realistically produce (mimic) the measured KM survival data from Section 3.2 and later. Figure 7 in Section 3.6 (in the main document) shows results from a "mixed scenario" simulation together with the corresponding measured results, for the age cohort that is the main focus of this paper,

the group born 1940-1950 (no WLZ long-term care). This is the scenario following from the specific parameter values shown in Table A.6. Such a mixed scenario with both a large transient healthy vaccinee bias component and a substantial vaccination misclassification bias component seems realistic, given all primary and secondary data and analysis presented above. It leads to a good "fit" between simulated and measured results. It is also "robust", in the sense that small variations in the underlying parameters can produce similar simulation results and similar good fits, suggesting that it is not very sensitive to precise parameter settings. All this is suggestive of realism with respect to real-world circumstances at that time. However, other, substantially different parameter settings can also lead to good and reasonable fits.

It is technically possible to obtain simulation results with a good fit using only limited vaccination status registration issues; but it requires an extreme HVE bias, corresponding to only very few vulnerable individuals having been vaccinated. A set of parameters for such a scenario are shown in table A.7. Figure A.22 shows the simulation results from this scenario. Such a scenario seems less realistic, given our own analyses of still many high-comorbidity individuals being vaccinated (see Section 3.5); public reports on their general prioritization and high vaccination rates [24, 37]; and independent reports based in part on separate (NIVEL) vaccination records confirming that many vulnerable and even end-of-life individuals were vaccinated according to GP records [19], despite many of those vaccination records missing in CIMS.

Table A.7: Quantitative Bias Analysis 1, example simulation parameters. These parameters are for the age cohort that is the main focus of this paper, the group born 1940-1950 (no WLZ long-term care). This particular set of parameters represents an "Extreme HVE" simulation (see text for explanation).

| | | **Vaccinated** | **Unvaccinated** |
|---|---|---|---|
| **Number of people** | N (%) | 300000 (75) | 100000 (25) |
| **Vulnerability group** | N (% in V/U group) | | |
| Low vulnerability | | 297640 (99.21) | 82400 (82.40) |
| Med. vulnerability | | 1960 (0.65) | 8000 (8.00) |
| High vulnerability | | 400 (0.13) | 9600 (9.60) |
| **Vulnerability group** | N (% in whole group) | | |
| Low vulnerability | | 297640 (74.4) | 82400 (20.6) |
| Med. vulnerability | | 1960 (0.5) | 8000 (2.0) |
| High vulnerability | | 400 (0.1) | 9600 (2.4) |
| **Vulnerability group** | probability death per day | | |
| Low vulnerability | | 0.00005 | 0.00005 |
| Med. vulnerability | | 0.006 | 0.006 |
| High vulnerability | | 0.15 | 0.15 |
| **Vulnerability group** | probability missing vacc. reg. | | |
| Low vulnerability | | 0.10 | n.a. |
| Med. vulnerability | | 0.10 | n.a. |
| High vulnerability | | 0.10 | n.a. |
| **Short-term special period** | duration in days | 28 | 28 |
| **Vuln. group, short-term:** | after death: extra prob. missing vacc. reg. | | |
| Low vulnerability | | 0.00 | n.a. |
| Med. vulnerability | | 0.00 | n.a. |
| High vulnerability | | 0.00 | n.a. |
| **Vuln. group, short-term:** | extra prob. death per day | | |
| Low vulnerability | | 0.00 | 0.00 |
| Med. vulnerability | | 0.00 | 0.00 |
| High vulnerability | | 0.00 | 0.00 |

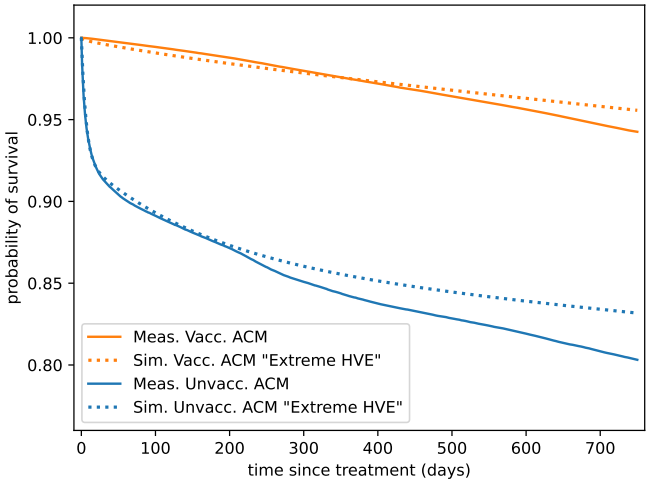



Figure A.22: Measured survival curves for the 1940-1950 age cohort (no WLZ, ACM), together with *simulated* versions, as produced by the mechanistic QBA model simulation. This particular simulation is based on an "Extreme HVE" scenario.

Table A.8: Quantitative Bias Analysis 1, example simulation parameters. These parameters are for the age cohort that is the main focus of this paper, the group born 1940-1950 (no WLZ long-term care). This particular set of parameters represents a "Covid high IFR" simulation (see text for explanation).

| | | **Vaccinated** | **Unvaccinated** |
|---|---|---|---|
| **Number of people** | N (%) | 300000 (75) | 100000 (25) |
| **Vulnerability group** | N (% in V/U group) | | |
| Low vulnerability | | 281200 (93.73) | 93867 (93.87) |
| Med. vulnerability | | 18400 (6.13) | 6000 (6.00) |
| High vulnerability | | 400 (0.13) | 133 (0.13) |
| **Vulnerability group** | N (% in whole group) | | |
| Low vulnerability | | 281200 (70.3) | 93867 (23.47) |
| Med. vulnerability | | 18400 (4.6) | 6000 (1.5) |
| High vulnerability | | 400 (0.1) | 133 (0.03) |
| **Vulnerability group** | probability death per day | | |
| Low vulnerability | | 0.00003 | 0.00015 |
| Med. vulnerability | | 0.001 | 0.0025 |
| High vulnerability | | 0.008 | 0.015 |
| **Vulnerability group** | probability missing vacc. reg. | | |
| Low vulnerability | | 0.10 | n.a. |
| Med. vulnerability | | 0.10 | n.a. |
| High vulnerability | | 0.10 | n.a. |
| | | | |
| **Short-term special period** | duration in days | 28 | 28 |
| **Vuln. group, short-term:** | after death: extra prob. missing vacc. reg. | | |
| Low vulnerability | | 0.00 | n.a. |
| Med. vulnerability | | 0.00 | n.a. |
| High vulnerability | | 0.00 | n.a. |
| **Vuln. group, short-term:** | extra prob. death per day | | |
| Low vulnerability | | 0.00 | 0.004 |
| Med. vulnerability | | 0.00 | 0.004 |
| High vulnerability | | 0.00 | 0.01 |

As a sanity check, we also experimented with a simulation scenario based on some but limited HVE and negligible vaccination status misclassification bias, together with high short-term mortality in the unvaccinated. A set of parameters for this scenario are shown in Table A.8. Such high short-term mortality exclusive to the unvaccinated would presumably be due a significant Covid wave; and it would simultaneously imply very high and immediate vaccine protection, such that vaccinated people are not affected by the Covid wave. This would be the kind of scenario compatible with reported conclusions on high Covid vaccine effectiveness following the first vaccination round in terms of all-cause mortality, implied in the earlier studies [18, 24, 19, 20, 21]. Figure A.23 shows the simulation results from this scenario.

To obtain these simulation results that have a reasonable fit to measured results for this 1940-1950 age group, this variant of the simulation model requires parameters equivalent to a situation where a big Covid wave infects almost everyone in the unvaccinated group within a period of 4 weeks or a little more. The mean Infection Fatality Rate (IFR) in that group over that period (in the spring of 2021) implied by these parameters (shown in Table A.8) is around 10% or more. Simultaneously, vaccine protection immediately after dose 1 would have to be almost perfect. And finally, it requires that most of those Covid deaths were classified as non-Covid deaths (compare with results discussed in Section 3.2 and Figure 3). All this is clearly unrealistic and incompatible with what we know from realistic assessments, both in our own study and elsewhere.

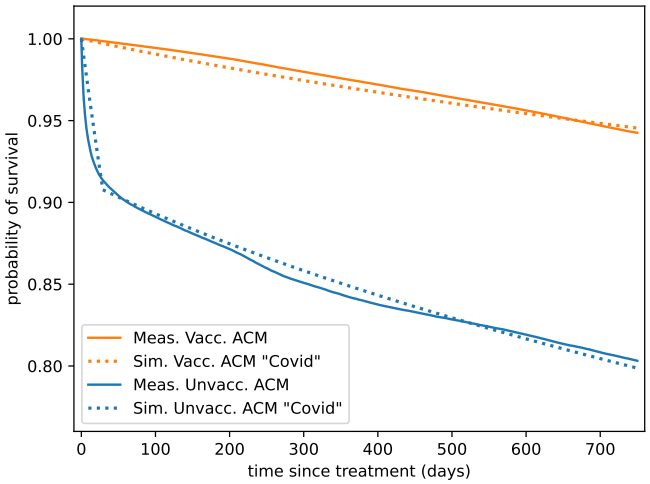



Figure A.23: Measured survival curves for the 1940-1950 age cohort (no WLZ, ACM), together with *simulated* versions, as produced by the mechanistic QBA model simulation. This particular simulation is based on a "Covid high IFR" scenario with no healthy vaccinee effect or substantial vaccination gaps but a big Covid wave, coinciding with the vaccination campaign and affecting the unvaccinated group very strongly.

*Appendix A.15. Quantitative bias analysis 2: Heuristic bias adjustment using negative control data*

The second QBA approach is complementary to the first QBA approach described in the previous section, Appendix A.14. It largely ignores the potential underlying factors or mechanisms driving the biases, modeled in the first QBA approach. In contrast, it is numerical and purely data-driven. It corresponds to a heuristic *negative control* data-based systematic bias correction that has strong similarities to several other QBA efforts that use negative control data in a similar way, both in the Covid domain and outside it [33, 50, 51, 52, 53, 49].

It is based on the core assumption that the Covid vaccines should not meaningfully reduce risk for causes of death other than Covid (the negative control). This assumption (or something very similar) is used also elsewhere in this paper, in other Covid studies, and in studies of other vaccines and viruses like influenza [25, 26, 54, 27]. In other words, non-Covid mortality data, and observed differences therein between vaccinated and unvaccinated groups, constitute reasonable negative control data, which may be exploited quantitatively in QBA.

Based on this starting point, the method estimates a time-varying multiplicative correction factor that effectively balances non-Covid mortality between the two groups. This corresponds to a simple normalization or scaling approach, but one where the multiplicative scaling factor changes over time (from start of follow-up of the measured non-Covid mortality/KM data). It then applies this correction factor to the measured all-cause mortality rates per day, which are accordingly normalized (and also Covid mortality rates, as a secondary target). Based on this, finally, bias-adjusted KM survival curves can be calculated.

The method starts by taking the death rates per day underlying the non-Covid Kaplan-Meier data (or, equivalently, deriving them from those KM data). These correspond to what is also called instantaneous hazard data per day, in epidemiology, which are a general basis for many epidemiological methods. We subsequently perform a light de-noising by smoothing those (noisy) death rate data. In our experiments, we do this by applying a simple (3-day) moving average filter, effectively using previous-day and next-day death rate together with current-day death rate to obtain a de-noised estimate of current-day death rate. These are applied to both the unvaccinated non-Covid mortality and the vaccinated non-Covid mortality.

The next steps can be operationalized in multiple ways. The simplest approach is to simply calculate the correction factor as the ratio of the unvaccinated non-Covid mortality and the vaccinated non-Covid mortality. All-cause and Covid adjusted mortality per day for the unvaccinated are subsequently normalized by simply dividing their (death rate) data per day by that correction factor (normalization factor) per day.

However, this ignores the fact that the vaccinated non-Covid mortality is *also* affected by the combined systematic bias effects (of HVE and misclassification bias), albeit to a lesser extent and in the opposite direction. Thus, the somewhat sophisticated, better and cleaner method is to adjust *both*, to an overall, (theoretically) unbiased estimate. We use this method for the results presented in this paper. This requires determining, in addition to the unvaccinated and vaccinated data, overall non-Covid mortality KM survival/death rate per day data, which represent the "average" unbiased combined mortality patterns. This is realized by taking a large enough, random and representative sample of individuals from the whole age group (that we wish to calculate adjustments for), and calculating their combined KM/mortality data from that.

For the 1940-1950 (no WLZ) age cohort, the focus of most of our analyses and also quantitative bias analyses, we take a sample of 400 000 individuals from the group of more than 1.2 million individuals. For the 1940-1950 (no WLZ) age cohort, the sample contains approximately 92% vaccinated individuals. Start of follow-up (necessary for KM survival analysis) is determined in essentially the same way as in the stratification and matching process (see Section 2.5 and Appendix A.5), and using the Prescription Time-Distribution Matching (PTDM) requirement [42]. Thus, follow-up for vaccinated individuals starts on their (first dose) vaccination dates. Follow-up for unvaccinated individuals start on dates Monte Carlo-sampled according to the real-world vaccination dates distribution; and requires that the unvaccinated person is alive at that date. Altogether this realizes overall mortality patterns that are a representative

"mix" of unvaccinated and vaccinated individuals from the age group, reflecting the real-world vaccination rate; with start dates of follow-up similar to and representative of what was used in all KM data analyses in this study.

Next, we use these overall non-Covid mortality data as the "basis" for the adjustment (see Figure A.24 for a schematic illustration). The corresponding per-day death rate data is similarly smoothed by applying a 3-day moving average filter (see Figure A.25). Then, for both the unvaccinated and vaccinated non-Covid mortality, we calculate correction factors $f$, time-dependent (from start of follow-up) and per-day, with respect to this overall per-day death rate:

$$f_u^{NC}(t) = \frac{d_u^{NC}(t)}{d_o^{NC}(t)} \tag{A.3}$$

and

$$f_v^{NC}(t) = \frac{d_v^{NC}(t)}{d_o^{NC}(t)} \tag{A.4}$$

in which:

1. $d_u^{NC}(t)$ is the Negative Control (in our case: Non-Covid) mortality death rate (per day, smoothed) of people in the Unvaccinated group over time t, in the measurement period since start of follow-up;
2. $d_v^{NC}(t)$ is the Negative Control (in our case: Non-Covid) mortality death rate (per day, smoothed) of people in the Vaccinated group over time t, in the measurement period since start of follow-up;
3. $d_o^{NC}(t)$ is the Negative Control (in our case: Non-Covid) mortality death rate (per day, smoothed) in the Overall group randomly sampled from and representative of the whole age (and WLZ) group, over time t, in the measurement period since start of follow-up;
4. $f_u^{NC}(t)$ is the time-dependent negative control-based correction factor (per day) for the Unvaccinated group in the measurement period since start of follow-up;
5. $f_v^{NC}(t)$ is the time-dependent negative control-based correction factor (per day) for the Vaccinated group in the measurement period since start of follow-up.

Adjusted ACM death rates per day for the Unvaccinated and Vaccinated groups are subsequently calculated by:

$$d_{u,adj}^{ACM}(t) = \frac{d_u^{ACM}(t)}{f_u^{NC}(t)} \tag{A.5}$$

and

$$d_{v,adj}^{ACM}(t) = \frac{d_v^{ACM}(t)}{f_v^{NC}(t)} \tag{A.6}$$

in which $d_{u,adj}^{ACM}(t)$ and $d_{v,adj}^{ACM}(t)$ are the Unvaccinated and Vaccinated *adjusted* per-day ACM death rates, respectively.

These adjusted death rates are subsequently used to calculate the adjusted ACM Kaplan-Meier survival curves. (And the same logic and formulas, the same proportional negative-control based correction factors but now applied to Covid death rates, is used to adjust the Covid KM data.)

For illustration, we again use the 1940-1950 (no WLZ) age cohort, the primary group focused on in this paper. The basic logic of the adjustment is visualized schematically in Figure A.24. Again: this corresponds to a normalization of non-Covid mortality for either the unvaccinated group or the vaccinated group towards the (theoretically) unbiased estimate of the overall non-Covid mortality (= negative control) KM data. And because in our case the (very heterogenous) unvaccinated group is affected most by the biases observed in the negative control data, correction for that group must be, and is, greatest. The vaccinated group is affected by the bias in the opposite direction, but its bias and corresponding correction is smaller.

The left plot in Figure A.25 shows the (lightly smoothed) non-Covid death rates per day for the unvaccinated, vaccinated, and overall groups (note the log scale); showing the disparities and bias measured/assumed in this QBA method. The right plot shows the calculated correction factors derived from this. The results, adjusted Kaplan-Meier curves for ACM and Covid mortality, are shown in Figure A.26. In the main text, in Section 3.7, we further discuss interpretation of these results.

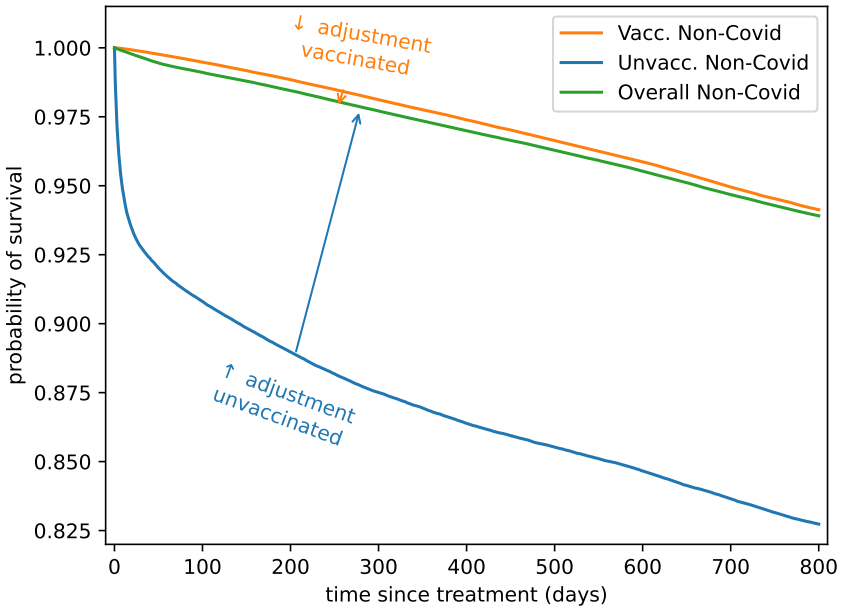



Figure A.24: Schematic illustration of the adjustment for vaccinated and unvaccinated individuals of age cohort 1940-1950, no WLZ. The adjustments are based on using measured vaccinated and unvaccinated non-Covid mortality data over time as the negative control data, and the adjustments for each are calculated as time-dependent proportional adjustments toward the unbiased overall esimate (green).

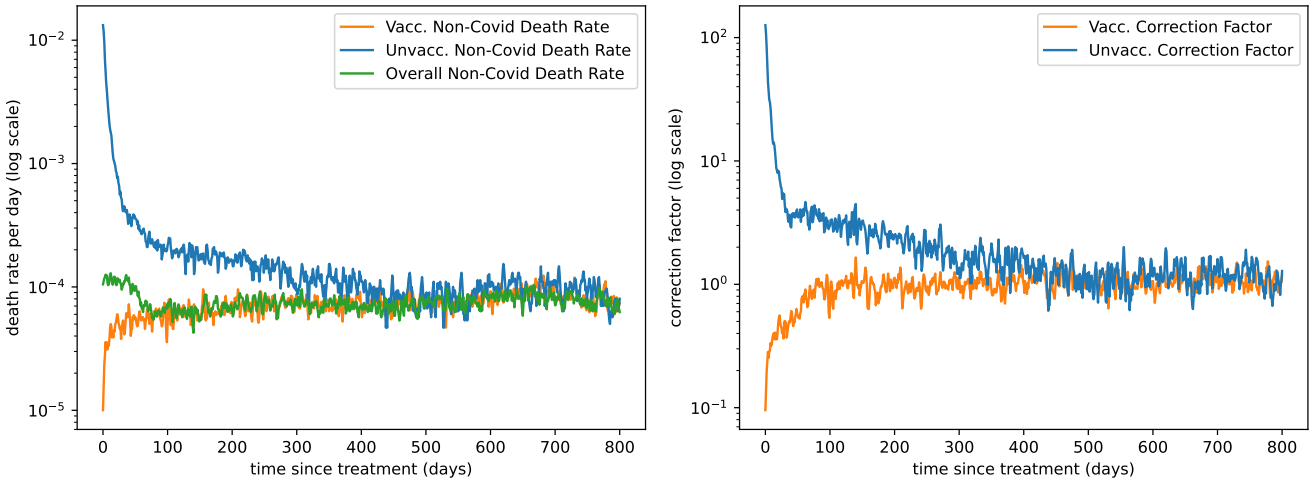



Figure A.25: Left: Death rates over time underlying the non-Covid mortality KM curves (lightly smoothed), for age cohort 1940-1950, no WLZ. Right: Correction factors over time derived from the ratios of the non-Covid death rates. Note the log scale on the $y$-axis.

In Figures A.27 and A.28 we show some results for both younger age groups and (on the other end of the spectrum) the oldest and most frail groups. For younger age cohorts (Figure A.27, age group 1960-1980), calculated bias adjustments are qualitatively similar. However, the remaining calculated

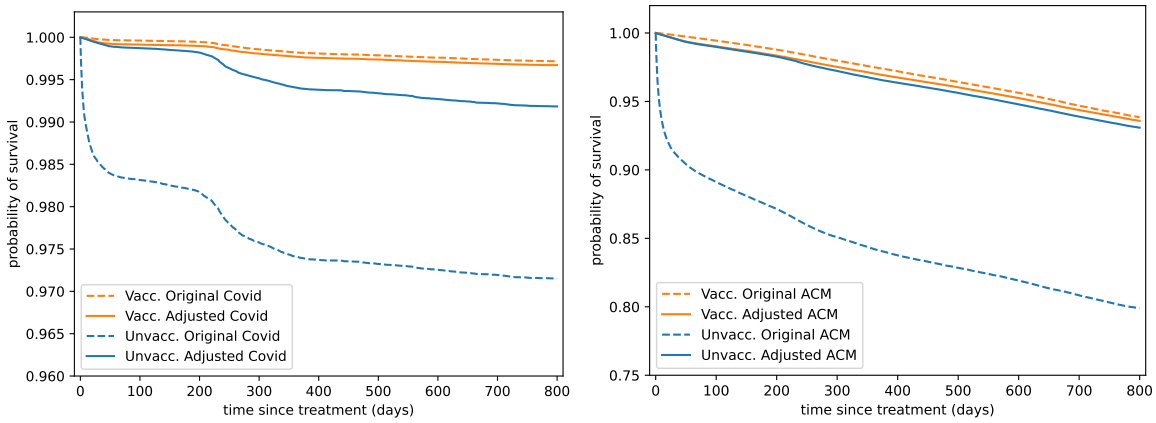



Figure A.26: Adjustment for vaccinated and unvaccinated individuals of age cohort 1940-1950, no WLZ. The KM survival data adjustment for mortality from Covid is shown in the left figure; the KM survival data adjustment for all-cause mortality (ACM) is shown in the right figure. Note the different scales on the $y$-axis.

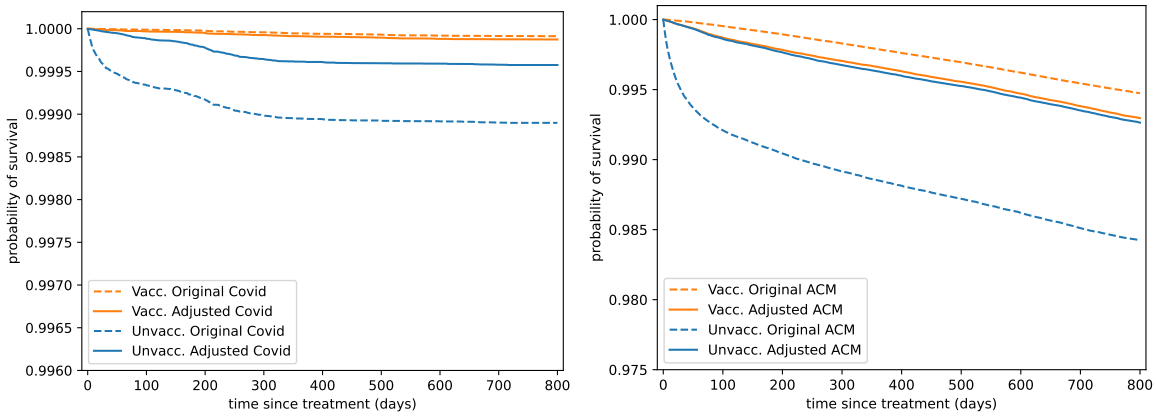



Figure A.27: Adjustment for vaccinated and unvaccinated individuals of age cohort 1960-1980, no WLZ. The KM survival data adjustment for mortality from Covid is shown in the left figure; the KM survival data adjustment for all-cause mortality (ACM) is shown in the right figure. Note the different scales on the $y$-axis.

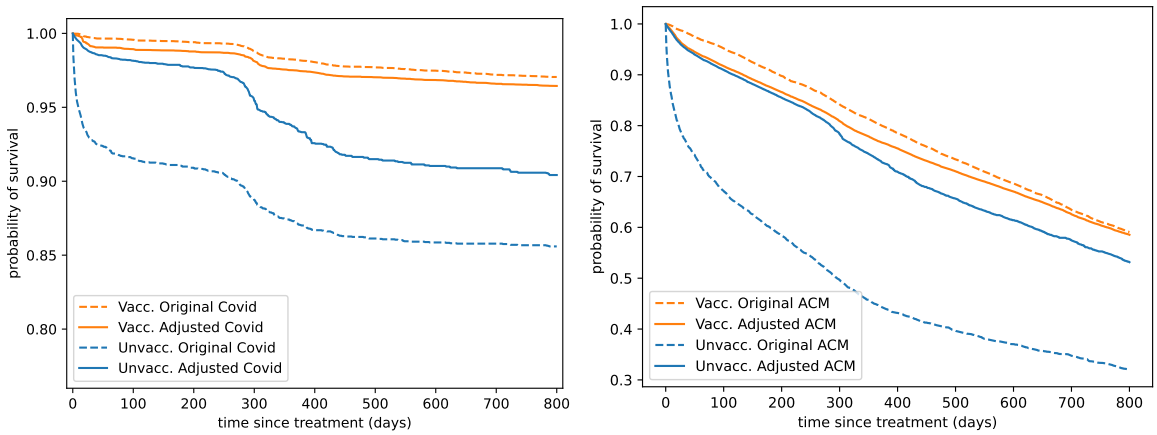



Figure A.28: Adjustment for vaccinated and unvaccinated individuals of age cohort 1940-1950, *with* WLZ. The KM survival data adjustment for mortality from Covid is shown in the left figure; the KM survival data adjustment for all-cause mortality (ACM) is shown in the right figure. Note the different scales on the $y$-axis.

vaccine protection becomes ever smaller, and quickly converges to zero for ever younger cohorts. For this 1960-1980 age group, in absolute terms (ARR), only an extremely small (and uncertain) protective effect of approx. 0.03 percentage point in all-cause mortality protection remains for the vaccinated group, after 800 days. For the oldest and/or most frail cohorts (Figure A.28, age cohort 1940-1950 *with* WLZ), substantial apparent vaccine protection after adjustment does remain; but the results are likely strongly affected by the limitations of the adjustment method. Again, the main text discusses these results, and the limitations, in more detail.

This QBA bias adjustment approach has the attractive property that it corresponds to a pragmatic, heuristic, data-driven, net correction on the all-cause (or Covid) KM data, without needing to disentangle the composite sources of bias or explicitly model the causal mechanisms. It only assumes that the net residual biases (HVE and vaccination status misclassification together) affect all-cause and Covid mortality proportionally to their observed effect on non-Covid mortality, the observed "pseudo-effect" on the negative control data, as it sometimes called [33].

Generally speaking we believe that the adjustment method produces reasonable adjustment results; but at the same time, the results are uncertain, and we must acknowledge limitations of the heuristic adjustment method, in either direction. The method depends on non-Covid mortality being clean and independent and suitable for this adjustment approach. More subtle healthy vaccinee or selection effects not captured by clear differences in non-Covid mortality cannot be resolved by this bias adjustment method, by definition. This could for example include immunocompromised patients who, for that reason, could not be vaccinated; and who are also extra susceptible to SARS-CoV2 infection and subsequent death. Furthermore, both SARS-CoV2 infection's acute or non-acute impact on non-Covid mortality *and* potentially any vaccine-associated negative impact on non-Covid mortality become invisible, using this bias adjustment method. Thus, if non-Covid mortality data contains relatively many deaths that were actually from Covid but that were misattributed [40, 62], then the bias adjustment from our method would be too large; and this would lead to underestimation of the "true" vaccine effectiveness profile. If, alternatively, the non-Covid mortality data contains relatively many deaths which can and should realistically be attributed (at least partially) to the Covid vaccines [24, 59, 61, 63], then the bias adjustment from our method would be too small; and this would lead to overestimation of the true vaccine effectiveness profile and safety profile.

Another possible objection to this method could be that, in a way, we appear to *build in* large adjustment of the ACM curves, by "piggyback riding" on the natural situation/observation that all-cause mortality is mostly composed of non-Covid mortality. In a way, the method does that. But importantly, we simply show what happens given that that observation and situation is the case. We observe that non-Covid mortality is by far the largest component of all-cause mortality. (This did not have to be this way; but it has turned out this way.) Next, we observe very large apparent, but counter-intuitive, "protection" of Covid vaccines against non-Covid mortality; which we therefore assume to result from a systematic bias effect. We also observe that the same time-varying bias effect appears to affect Covid mortality proportionally, at least to a large extent. We then simply calculate what happens if we follow these observations and assumptions, and make them precise in a mathematical operationalization. Thus, in other words, this QBA method calculates and makes precise what happens if one follows the intuition that apparent non-Covid mortality protection from the vaccines must largely be bias-driven.

In summary, this bias adjustment does not claim perfect accuracy, but it makes precise what happens in terms of adjusted KM survival data if one accepts non-Covid mortality as a good and reasonable negative control, and if one accepts a straightforward multiplicative (proportional) correction factor based on that. In a way, it logically follows time-dependent proportional hazards logic, a key staple of epidemiological statistical methods.

In any case, we believe that these bias-adjusted results, while imperfect, are probably closer to the "true" vaccine effectiveness profile (whatever that is) than what is obtained from both our own direct primary analyses and those of previous studies based on the same dataset [18, 24, 19, 20, 21], which largely ignored, underestimated, or minimized these large systematic biases.